Land Reform as a Panacea? Evidence from Rural Protests in Latin America[1]

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Abstract

This paper investigates the efficacy of major land reform implementations in alleviating large-scale rural protests in Latin America. Our analysis harps on a novel channel involving church or clergy induced democratic participation given the fact that religion plays very important role in Latin American countries. The empirical strategy uses both short-run flexible event-study analysis as well as instrumental variable estimation to overcome endogeneity concerns. We find robust and consistent support for redistributive and expropriative type reforms as statistically significant pacifiers for rural protests. The novel channel variable., *viz.*, the church or clergy induced democratic participation provides positive impetus to land reform implementations in the first stage instrumental variable analysis. The amount of land redistributed due to the reforms and associated decline in the land inequality patterns (measured using land Gini coefficients) are identified as two plausible mechanisms to explain the baseline findings.

JEL codes: D74, O11, O54, Z12.

Keywords: land reform, rural protests, church, democratic participation, redistribution, expropriation.

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1. Introduction

This paper investigates a particular development policy involving land reforms and their possible mitigation effect on large-scale rural protests in Latin America. Our focus is on the long-run impact of various land reform implementations in the 20th century (1900-2010, from Bhattacharya et al., 2019) and their plausible beneficial effect in terms of reducing massive rural protest incidents over a long-run period (1930-2008, from Albertus, 2015). In so doing, we focus on a novel mechanism involving democratic participation through Church or clergy-initiated activities like years of active Base Christian Communities and decipher their possible soothing impact on rural protests. We assemble a unique country-level dataset documenting the active years of clergy or Church induced democratic participation in Latin America[2] and analyse how such participatory activities could be linked with more land reform implementations which in turn diminishes the likelihood of large-scale protests.

Protests or demonstrations are quite prevalent in the world as evidenced in the recent large-scale protests in the USA for racial justice and in Sri Lanka due to economic mismanagement. The Carnegie Endowment for International Peace (carnegieendowment.org) documents that as of July 2022, more than 230 significant antigovernment protests have erupted in 110 different countries in the world. Machado, Scartascini and Tommasi (2011) link protests in seventeen Latin American countries with weak institutions. Acemoglu and Robinson (2012) and Aidt and Frank (2015) provide evidence on how political protests drive economic, social and political change over centuries.

In the same vein, Albertus (2015) documents major rural protests in a number of countries in Latin America for the period 1930-2008. These demonstrations could be broadly categorized into the forms of land invasions (Dominican Republic, El Salvador, Honduras, Nicaragua, Panama, and Venezuela), strikes (Argentina, Bolivia), protests (Chile, Paraguay), demonstrations (Bolivia, Mexico), blockades (Ecuador), movement of landless rural workers and farmers (Brazil) as well as rural insurgency (Colombia, Cuba, Guatemala, Mexico, Peru, Uruguay). In terms of magnitudes, these demonstrations would in general, be termed as low magnitude incidents, as except for rural insurgency, the likelihood of human casualties is quite low. However, it is important to note that these protests could be catalysts to subsequent high intensity conflicts like guerrilla insurgencies as documented by the outcomes of such rallies in Albertus (2015, Table 2.3). Thus, even if these are low-intensity incidents, their possible long-term ramifications would be very significant. Hence, it is of first order importance to analyse if such major protests could be pacified with pertinent policy intervention.

We take one such development policy tool involving land reforms, and in particular, the cumulative impact of such reform implementations and investigate how these reform implementations could act as pacifiers in diminishing protests. Land reform itself is a very important issue from the perspective of Latin America given the prevalence of large amount of land concentration due to the historical agrarian structure evolved through the colonization period (Sokoloff and Engerman, 2000). Albertus and Kaplan (2013) investigate whether land reform could be used as a strategy for reducing civil conflict in Colombia. Similarly, Albertus (2020) shows that greater land reform in Peru reduced conflict by augmenting retaliation

[2] The Data Appendix provides detailed narrative of the dataset.

capacity akin to improving state capacity in the civil conflict literature (Blattman and Miguel, 2010).

Given that the land reform episodes are major milestones in countries long-run trajectory (Bhattacharya et al. 2019; Albertus, 2015), it would be important to analyse the long-term consequence of such reforms on major protests witnessed in Latin America. Note that land reforms are not uniformly implemented in a number of countries in Latin America, for instance, in Argentina (Bhattacharya et al. 2019). In addition, even if there are multiple land reforms enacted over the years, they are always partially implemented as in Costa Rica, Guatemala and Honduras. Then in a number of other countries like Belize, Bolivia and Colombia, some particular land reforms are fully implemented, and others are either partially implemented or not implemented at all.[3]

It is interesting to note that the land reforms are associated with a number of different objectives. We exploit these various motives of land reform implementations relevant for Latin American countries from Bhattacharya et al. (2019). These are: (i) land reform implementation as a package subsuming all different strands of reforms as illustrated in (ii) onward, (ii) reforms mandating upper limits in landholding or land ceiling, (iii) land expropriation motive where excess land over the mandated ceiling have been taken over from landlords for later reallocation to land-poor or landless people, (iv) redistribution motive where lands were redistributed to landless along with full end-user rights on land, (v) a combination of land ceiling, land expropriation and land redistribution motives of land reforms as often expropriation and redistribution are implemented together in a number of countries in Latin America, (vi) pro-poor land reforms where reforms are specifically targeted towards the poor or landless or land-poor or squatters as mentioned in the reform initiatives, so this can be treated as a package explicitly designed for the poor, (vii) tenure security improvement as the sole reform motive where the land tenure arrangements are improved without transferring physical lands and end-user rights to land, and (viii) a combination of land reform motives where end-user rights in land are given, viz., (a) consolidation type reforms where land is mainly re-parcelled into unitary blocks for improved use with end-user rights in land granted; (b) distribution of land from state-owned farms to citizens; and (c) privatization or sale of state-owned farmlands to individuals.

In addition, the following land reform motives are taken from Albertus (2015). First, we use the global major/minor land reform with the major "*is defined as the expropriation and redistribution of at least 10 percent of cultivable land over a continuous period with at least one year in which more than 1 percent of cultivable land is expropriated.*"[4] Minor reforms are depicted as "*those where there was expropriation and redistribution as part of a land reform program but at a threshold short of major reform.*" The second motive is the global national land reform from Albertus (2015). Note that these motives are distinct from the motives taken from Bhattacharya et al (2019).

The Church induced democratic participation, the new mechanism proposed in the paper stems from the fact that Church or clergy plays a very important role in Latin America in shaping up the social impact on development in general through civil society engagement

[3] Please refer to the Supplementary Appendix of Bhattacharya et al. (2019), and especially Table A1 in regard to major land reforms enactment bills and their implementation status.
[4] See notes of Albertus (2015), Table 8.1, page 273.

(Smith 2017; Linz and Stepan, 1996). Given the prominence of religion in Latin America, we investigate a channel of clergy induced democratic participation (CIDP) including years of or active years of base Christian communities (known as *Comunidades Eclesiales de Base* or CEB) which may have a bearing on land reform decisions and subsequent amelioration of low-intensity conflict manifested by large scale rural protests. These clergy-initiated activities span the 'broader special public role' dimensions which could be listed as 'human rights monitoring, organizing labor protests, establishing cooperatives and credit unions, providing health care and education, fundraising, supporting solidarity campaigns, publishing and other reform activities' among others (Jordan 2022; CIA Report 1986).

In the empirical analysis, first we analyse whether the land reforms would act as deterrents to rural protests by looking at the implementations of reforms within a short time span using a flexible event study analysis as in Bhalotra et al. (2019). Here we employ land reforms implemented as treated and no reforms implemented as control groups within a 10-year before and after time window. Even within this short time span, we find some support that land reforms remain effective as a protest mitigation mechanism. However, this type of investigation would treat the reforms as purely exogenous which may not be the case always as protests could be initiated due to lack of reforms in some instances (Albertus, 2015; de Janvry et al, 2015). Hence, endogeneity due to reverse causality would threaten our identification strategy.

Thus, to address such concerns, we resort to instrumental variable estimation. One plausible exogenous factor affecting the land reform implementations will be the impact of elite split as documented in Albertus (2015). This spilt is between landed elites and political elites where landed elites have wealth and power based on vast amounts of lands at their disposal. The political elites composed of military and civilian politicians and have the "power and organizational capacity to run the government" (Albertus, 2015, data appendix pg. 6). Based on Figure 5.2 (pg. 153, Albertus 2015) and the narrative in the data appendix of Albertus (2015), we code the years of elite split and use this as an exogenous instrument for land reform implementation decisions.

Albertus (2015) demonstrates that elite split has important positive effect on land reform decisions. Given the focus of the land reforms are key milestones in a country's long-run trajectory (Bhattacharya et al. 2019), hence these big-ticket events could be linked with the power struggles between political elites and landed elites in Latin America. Albertus (2015) documents how elite split would hasten land reform decisions based on the intuition that such reforms help consolidating the support of the stakeholders like peasants, poor people, unemployed, union members etc to remain in power. This particular characteristic of elite split makes it the relevant instrument to be used in the identification purpose in the econometric analysis. In addition, these divisions did not instigate any major rural protests (which is the dependent variable or outcome variable in the empirical analysis), and hence, excludability restriction in terms of plausible instruments should also be satisfied.

Our endogeneity corrected results reveal that cumulative land reform implementations reduce protests in general. There is quite a bit of heterogeneity in regard to specific land reform motives. Amongst the reforms, the strongest mitigating impact is attributed to the redistributive and expropriating type reforms individually as well as the combination of such reforms implemented jointly. In addition, we find that tenure security improvement type

reforms which does not entail physical transfer of land and focus on enhancing tenure arrangements between land tillers and landowners also has a detrimental impact on large-scale rural protests. The main results survive a number of robustness checks involving different empirical estimation techniques and falsification exercise with alternative outcome variable from the extant literature.

The novel control variable used in the study, *viz.*, the church or clergy induced democratic participation (CIDP), provides positive impetus to land reform implementations in our first stage instrumental variable analysis. Consequently, the coefficients of CIDPs become negative in the second stage instrumental variable regressions, lending credence to the idea that active years of democratic participation also have a direct and soothing impact on large-scale rural protests in addition to their indirect effects through enticing reform implementations at the first stage. In the paper's context, the CIDP heralds in a new democratic window of opportunity.

In regard to mechanisms, we pinpoint two plausible pathways involving proportion of land redistributed from Albertus (2015) and reduction in land inequality using data from Frankema (2010). First, there is strong evidence that land reforms implemented redistributed lands to the needy (could be interpreted as improving extensive margins in land holdings), thus creating more productive income earning opportunities, which in turn reduced rural protests. Second, the land inequality gini coefficients show statistically significant reductions in land holding distributions, akin to an improvement in intensive margin of land holding patterns. In line with the main results, the redistributive and expropriation type reforms as well as their combinations are associated with both higher proportions of land being redistributed as well as more reductions in land inequalities. Hence, these motives of land reform implementations are attributed to higher protest mitigations in the main analysis.

Our study is related to the following strands of literature. First, there is a body of scholarship investigating whether land reforms could be used as mitigation mechanisms to reduce civil strife in a number of countries in Latin America (Albertus and Kaplan, 2013; Albertus, 2019). We contribute to this strand by analyzing rural protests akin to low-intensity conflicts within the region over a long period of time. The investigation reveals that major land reform implementations, taken cumulatively and concentrating on expropriation and redistribution of land could be used as a pacifier in dousing such low-intensity civil strife.

Second, a number of papers highlight church-induced democratic participation and their impact on development in general through civil society engagement (Smith 2017; Linz and Stepan, 1996). Given the prominence of religion in Latin America, we investigate a channel of clergy induced democratic participation (CIDP) including years of or active years of CEB which may have a bearing on land reform decisions and subsequent amelioration of low-intensity conflict manifested by large scale rural protests. Our results show that CIDP hasten the reforms decisions in the first stage analysis, and especially for land redistribution mechanisms.

Third, the paper is, in a broader sense, related to studies like Caicedo (2019) and Castello-Climent, Chaudhary and Mukhopadhyay (2018) which show how missionaries helped in creating human capital in Latin America and India through establishing missionary schools. The present study's focus is on the clergy initiated democratic involvement which could be

treated as a complementary public good to formal education incentives and having a different effect in mitigating small-scale skirmishes.

The rest of the paper is organized as follows. In the following section, we briefly describe the connection between Church or clergy induced democratic participation and rural protest in Latin America. Section 3 outlines the empirical strategy including the data used. Section 4 presents the results and discussions. The mechanism results are presented in section 5. Finally, section 6 concludes the paper. In Appendix Tables section, we provide detailed empirical results from main tables in the manuscript. The Data Appendix sheds light on the data and coding decisions for the new channel variable, church or clergy-induced democratic participation.

2. Church or clergy initiated democratic participation and rural protests in Latin America

We focus on democratic participation through Church or clergy-initiated activities like years of active Base Christian Communities (*Comunidades Eclesiales de Base* or CEB) given the fact that religion plays an important role in Latin America (Smith, 2017; Bruneau, 1980). For example, the latest Pew Report of Religion in Latin America (2014)[5], shows that 69% adults are Catholic and 19% are Protestant when 30,000 respondents were surveyed in 2013-2014. Interestingly, close to 50% of both Christian faiths support charity work for the poor and around one-fifth of the Catholic respondents reported that they would persuade the government to help the poor.

Democratic participation through religious activities is an important dimension as there is a well-established literature in political science which show how religion in general and Catholicism in particular as well as politics are interrelated in Latin America (Sanders 1982). Sanders (1982) reviews three books on the subject and points to heterogeneities across countries on the issue of Catholicism and politics. In particular, Sanders (1982, pg. 256) writes, "*The church, to be sure, defines a special public (and political) role for itself. One dimension involves the denunciation of injustice and exploitation and the promotion of a better society through several strategies: statements analyzing the problems and sketching general directions for society, the formation of CEBs and other organizations where participants can develop a social orientation based on Christian presuppositions, encouragement of the laity to be active politically, and church activities such as defense of human rights and the poor under authoritarian and oppressive systems, especially where other institutions cannot function.*"

We exploit the 'broader special public role' dimension with democratic participation through church-initiated activities which would have a bearing on pacifying protests. These activities could be described as 'human rights monitoring, organizing labor protests, establishing cooperatives and credit unions, providing health care and education, fundraising, supporting solidarity campaigns, publishing and other reform activities' (CIA Report 1986). Thus, years of such participatory actions at the grassroot level could be seen as an impetus to significant policy decisions such as land reforms. Bruneau (1980) mentions how involvement in such grassroot organizations like the CEB in Brazil and other Central American countries are driving the local development agenda including establishment of production and consumer

[5] https://www.pewforum.org/2014/11/13/religion-in-latin-america/

cooperatives and hiring lawyers to challenge arbitrary political and administrative measures. The CIA Report (1986) documents active role plays by CEBs in (i) Nicaragua in the 1970s "*to promote social action and apply pressure to Somoza government*", (ii) El Salvador in the 1980s to give support to democratic reform initiatives, and (iii) Guatemala in the 1970s in the wake of widespread human rights abuse by the successive oppressive military regimes.

One empirically measurable aspect of local participation in religious activities is the advent and effective duration of CEBs which were represented by mainly the poor and non-elitist people under the aegis of sympathetic bishops within the Catholic Church.[6] The effectiveness of CEBs were determined by the local clergy or bishops pivotal in organizing such communities at the diocesan levels as happened in a number of Latin American countries like in Nicaragua and Honduras. The CEBs have certain other unique characteristics such as (a) these were formed at the local levels and lead by lay people or the masses (and not elites), (b) the CEBs were institutionalized after the conclave of Catholic Church in 1968 at Medellin, Colombia and the initial efforts were made in earlier conference of Catholic Bishops and under the directive of Catholic Church in between 1962 and 1965; thus making the advent of CEBs exogenous to the large scale rural protests,[7] (c) the effective diocesan clergy were primarily from outside the domicile countries and (d) the CEBs were free from direct influence of other institutional directives like forming coalition to defend the spread of communist influence in 1965.

A number of recent studies highlight how religious participation and democratic activities are interlinked with church participation treated as a broader civil society engagement (Vasquez and Williams, 2005; Smith, 2017). Both in developed countries like the USA (Verba, Nie and Kim, 1978; see also the classic study by Tocqueville, 1838) and developing countries like Chile (Bauwens and Lemaitre, 2014), church influenced social and democratic participation helped the needy, especially the poor and shaped subsequent development in a number of countries like the Democratic Republic of Congo (Kyamusugulwa and Hilhorst, 2015) and Guatemala (Wydick, Karp and Hilliker, 2011).

Based on the discussion above, we also include additional markers of clergy directed or church induced democratic participation. First, following Wang (2021), we code the years of radio address by prominent local clergies eliciting participation in democratic activities as in Honduras where one of the progressive bishops, Mr. Evelio Dominguez started vigorous rural engagement through regular 'radio talks' from 1961 onward (Shepherd 1993). Second, publication of statements or newspaper articles soliciting sympathizers for social activities like in the Dominican Republic in the 1970s are also coded as democratic participation years (US Country Studies for Dominican Republic, 1989). Finally, if local clergy themselves participated in some social activities like joining labor unions as happened in Costa Rica

[6] Note that the land reforms in the dataset do not include forcible expropriation of Church owned lands and redistributing it to the rural poor as happened in a number of Latin American countries (Mexico, 1857; Ecuador 1907) primarily in the late 19th century or early 20th century (Albertus 2015, pgs. 53-54). Hence the establishment of CEBs could not be treated as a reaction to such land reforms.

[7] There are instances that clergies directing CEBs were involved in rural protests in countries like Nicaragua, Guatemala, El Salvador, Brazil, Chile (CIA Report 1986) as well as Honduras (Shepherd 1993). Eventually some of these active clergies formed political parties and joined the political process like the Christian Democratic Party in Guatemala.

during the time of Archbishop Sanabria's tenure[8] (Brett, 1993), we also include these activity years within the realm of democratic participation as exemplars.

3. Empirical strategy and data

3.1. Event study analysis

We start with the event study analysis which takes the following form:

$$Lsrp_{it} = \alpha + \sum_{j=2}^{10} \beta_j^{lead}\, lr_i \times 1\{lead_t = j\} + \sum_{m=0}^{10} \beta_m^{lag}\, lr_i \times 1\{lag_t = m\} + \boldsymbol{\delta}.\boldsymbol{Z} + \varphi_i + \sigma_t + \varepsilon_{it} \quad (1)$$

where rural protests is the outcome or dependent variable taken from Albertus (2015). These protests occur at different time epochs and could be treated as incidents of protests[9] akin to the incidents of civil conflict (Hull and Imai, 2013; Fearon and Laitin, 2003). The rural protest incidents are denoted by '*Lsrp*'. '*i*' stands for country and '*t*' refers to years. It is interesting to note that these large-scale rural protests lead to further civil conflict over time in a number of countries (Albertus, 2015). Thus, these rural demonstrations could be interpreted as low-intensity conflict incidents.

The above is a flexible event study setting as in Jacobson, LaLonde and Sullivan (1993), Stevensons and Wolfers (2006) and Bhalotra, Clarke, Gomes and Venkataramani (2019). The flexibility is in relation to the specification addressing different start time of land reform implementations in different countries. This is similar to the spirit of Goodman-Bacon (2018).

The main independent variable is denoted by *lr*, which represents land reform implementation at country *i*. *lr* takes the value of 1 if a country ever implemented a particular kind of land reform, and using the event study design, we interact this land reform with a set of leads and lags based on the year the land reform was implemented. We use 10 periods for leads and 10 periods for lags keeping in mind that these major land reform implementations are not occurring very frequently, and, thus, deciphering the impact of a particular land reform implementation within a period of 10 years before and 10 years after will be a reasonable timeframe to analyse its impact on the outcome variable.[10] In the model, we are also using appropriate controls as outlined later. In addition, we also control for unobservable factors using country fixed effects (φ_i) and year fixed effects (σ_t).

The coefficients of interest will be the estimates of β^{lead} as they would capture the impact after land reform implementations. The lag variables could check and test for any pre-trends which are very important in the underlying identification assumption of no parallel pre-trends (Bhalotra et al., 2019).[11] Freyaldenhoven et al. (2019) point that the absence of such pre-trends will be taken as a test of exogeneity of the policy variable, which is the land reform implementation in our context (described in detail in the next paragraph). Note that this test would be judged as partial as it can only identify parallel pre-trends but could not actually test

[8] 1940-1952. Please refer to country-specific data details in Data Appendix.
[9] The dependent variable takes the value 1 for years of protests, and 0 with years of no protests as documented in Albertus (2015).
[10] Note that individual country level studies by Albertus and Kaplan (2013) and Albertus (2020) also use similar time spans.
[11] The first lead is omitted as the baseline category.

for counterfactuals (Bhalotra et al., 2019; Kahn-Lang and Lang, 2018). The *a priori* expectation is that the overall impact of land reform implementations within the post 10-year window would reduce the likelihood of protest incidents.

We exploit various motives of land reform implementations from Bhattacharya et al. (2019) which are relevant for Latin American countries. These are: (i) land reform implementation as a package subsuming all different strands of reforms as illustrated in Bhattacharya et al. (2019), (ii) redistribution motive where lands were redistributed to landless along with full end-user rights on land, (iii) a combination of land ceiling, land expropriation and land redistribution motives of land reforms as often these motives are implemented together, (iv) pro-poor land reforms where reforms are specifically targeted towards the poor or landless or land-poor or squatters as mentioned in the reform initiatives, so this can be treated as a package explicitly designed for the poor and (v) tenure security improvement as the sole reform motive where the land tenure arrangements are improved without transferring physical lands and end-user rights to land,

In addition, the following land reform motive is taken from Albertus (2015), termed as global major/minor land reform, which "*is defined as the expropriation and redistribution of at least 10 percent of cultivable land over a continuous period with at least one year in which more than 1 percent of cultivable land is expropriated.*"[12] Minor reforms are depicted as "*those where there was expropriation and redistribution as part of a land reform program but at a threshold short of major reform.*"

A number of relevant control variables (denoted by the vector $\boldsymbol{Z}$) are used based on the extant literature (Blattman and Miguel, 2010; Esteban et al. 2012). These include time-varying controls like one year lag of log GDP per capita, log of population, democracy indicator[13]as well as an oil exports indicator dummy[14]which may have any impact on the large-scale rural protests have also been controlled for.

The novel variable, democratic participation through Church or clergy-initiated activities (CIDP) is also controlled for in the empirical analysis given their possible influence on rural protests. Moreover, time-invariant variables like non-contiguous territory and ethno-linguistic fractionalization as in Fearon and Laitin (2003) have also been controlled for. The standard errors are robust and clustered at the country level following the suggestion of Bertrand, Duflo and Mullainathan (2004).

The same event study approach is also used as the first checkpoint for placebo results with an alternative indicator of civil strife incidence used in the literature (Hull and Imai, 2013)) as outcome variable. Similarly, possible mechanisms (detailed later) are also analyzed with the above event study setting as the first touch point in the empirical analysis.

3.2. Baseline regression analysis

Then, we carry out the baseline analysis with large scale rural protests as the outcome or dependent variable using the following econometric specification:

[12] See notes of Albertus (2015), Table 8.1, page 273.
[13] Please refer to the Data sub-section below for detailed description of the variable used.
[14] Please refer to the Data sub-section below for detailed description of the variable used

$$Lsrp_{it} = \propto_{1i} + \beta_{1t} + \gamma.\text{tr} + \delta_1.LR_{it} + \boldsymbol{\theta_1}.\boldsymbol{Z} + \varepsilon_{1it} \quad (2)$$

The above empirical specification could be termed in the spirit of a two-way generalized difference-in-difference analysis as in Borusyak et al. (2022). The two-way effects are country-specific fixed effects ($\propto_{1i}$) and time effects (β_{1t}). Given the long-time span of the main explanatory variable (detailed description below) and the outcome variable (1930-2008), we also control for country-specific time trends in the above specification ($\gamma.\text{tr}$).

In this specification, the main explanatory variable is the cumulative effect of land reform implementations, denoted by *LR*. Given that the land reform episodes are major milestones in countries long-run trajectory (1900-2010, Bhattacharya et al. 2019; Albertus, 2015), it would be important to analyse the long-term consequence of such reforms on major protests witnessed in Latin America. Note that land reforms are not uniformly implemented in a number of countries in Latin America, for instance, in Argentina (Bhattacharya et al. 2019). In addition, even if there are multiple land reforms enacted over the years, they are always partially implemented as in Costa Rica, Guatemala and Honduras. Then in a number of other countries like Belize, Bolivia and Colombia, some particular land reforms are fully implemented, and others are either partially implemented or not implemented at all.[15] Thus, in the realm of difference-in-difference analysis, reforms which are fully or partially implemented could be used as the 'treated' group and the 'control' group will be reforms which are not implemented at all.

Under the reform category, we have analyzed various motives of land reform implementations as outlined before in the event study analysis. In addition, the following motives of reforms from Bhattacharya et al. (2019) are also incorporated. These include, (i) reforms mandating upper limits in landholding or land ceiling, (ii) land expropriation motive where excess land over the mandated ceiling have been taken over from landlords for later reallocation to land-poor or landless people, and (iii) a combination of land reform motives where end-user rights in land are given, viz., (a) consolidation type reforms where land is mainly re-parcelled into unitary blocks for improved use with end-user rights in land granted; (b) distribution of land from state-owned farms to citizens; and (c) privatization or sale of state-owned farmlands to individuals. Moreover, the global national land reform from Albertus (2015) is used as an additional motive of land reform. We expect that land reform implementations, taken cumulatively would have a negative bearing on large scale rural protests.

In line with the event study analysis, we have also used the same set of controls in equation (2), denoted by the vector **Z**. The standard errors are now robust to heterogeneity.

Given the nature of the outcome variable with the protests taking values of 1 and 0 interchangeably, we can treat the rural protests to be happening with some probabilistic nature. Thus, we validate the robustness of baseline ordinary least squares results using Poisson model with fixed effects as an alternative estimation strategy.[16]

[15] Please refer to the Supplementary Appendix of Bhattacharya et al. (2019), and especially Table A1 for information about major land reforms enactment bills and their implementation status.

[16] Note that negative binomial regression is another plausible model for checking robustness, however, we cannot estimate the negative binomial model incorporating fixed effects.

Note that the event study analysis and the baseline regression model presented above do not address possible endogeneity issues which are discussed below.

3.3. Addressing endogeneity with instrumental variable estimation

A number of plausible sources of endogeneity have to be accounted for in order to achieve identification. The main concern is that land reform implementations themselves could be a bone of contention or protests for stakeholders (Albertus and Mendalo, 2012; de Janvry et al. 2015), leading to possible reverse causality and biased estimates. The other sources of endogeneity related to plausible model misspecification and omitted variables are of lesser concern given that we have controlled for a number of fixed effects, trends as well as additional time-varying and time-invariant factors which may instigate civil strife based on the extant literature.

We use an instrumental variable strategy to address endogeneity concerns following the extant literature (Miguel, Satyanath and Sergenti, 2004; Blattman and Miguel, 2010). The instrumental variable is the years of elite split as measured by Albertus (2015). Albertus (2015) demonstrates that years of split between landed elite and political elite in the political hierarchy in Latin American countries could be used as an important driver of land reform decisions in Latin America. The implementations could be viewed as political economy verdicts with many different stakeholders like democratic/autocratic/military governments, elite citizens, rural masses, as well as urban groups involved in the reform and implementation initiatives (Bhattacharya et al, 2019, Albertus, 2015). Thus, years of elite split could be treated as relevant political economy instruments in line with the approach of Besley and Case (2000). Besley and Case (2000) employ an instrumental variable for a political confounder, similar to the present setup. We re-estimate the baseline specification in equation 2 using the instrumental variable two stage least squares method and this becomes our main specification in the empirical analysis.

Looking at the large scale rural protests from Albertus (2015), which could be broadly categorized into the forms of land invasions (Dominican Republic, El Salvador, Honduras, Nicaragua, Panama, Venezuela), strikes (Argentina, Bolivia), protests (Chile, Paraguay), demonstrations (Bolivia, Mexico), blockades (Ecuador), movement of landless rural workers and farmers (Brazil) as well as rural insurgency (Colombia, Cuba, Guatemala, Mexico, Peru, Uruguay), there is no evidence that the protests are linked with elite split. Similarly, focusing on the broad outcomes of such protests listed in Albertus (2015), there is no link that such demonstrations lead to split between political elites and landed elites in Latin America. Thus, years of elite split should be safely categorized as one pure exogenous variation affecting land reform decisions and orthogonal to the rural protests. Thus, exclusion restriction should be treated as passed comfortably when employing years of elite splits as instruments for land reform implementations.

In addition, we have performed placebo checks using instrumental variable estimation with conflict incidence as the outcome variable in equation 2. The key idea is that cumulative land reform implementations would reduce large-scale rural protests, and not necessarily conflict incidences as such conflicts could be linked with a number of reasons apart from rural protests or land reforms. Hence, we expect the signs of cumulative land reforms to be either insignificant or positive in the placebo check results.

The extant literature on civil conflict identifies income proxied by the level of GDP per capita or growth of GDP per capita as the main source of endogeneity (Blattman and Miguel, 2010; Verwimp et al. 2019). To address such endogeneity concerns, researchers use instrumental variables estimation with rainfalls or other measures as instruments for income, especially in the context of sub-Saharan African countries (Miguel et al. 2004; Ciccone, 2011). In this paper's context, such concerns are futile given that the outcome variable is rural protests which are not driven by income concerns. Even if there are some plausible issues that per capita income could still be endogenous to rural protests, in the empirical analysis, we use one year lagged value of GDP per capita in log terms as in Fearon and Laitin (2003) to address such concerns that income per capita may be correlated with major rural protests. Note that Albertus et al. (2018) also employ one year lagged independent variables.

In addition, we cross-checked our results excluding per capita income as a control in the baseline IV specifications and find that the main results still survive this robustness check.

3.4. Mechanisms

In order to decipher possible channels behind the main findings, we resort to two different pathways using the instrumental variable setting. The first one sheds light on the amounts of land redistributed as a result of implemented reforms. The idea is that with lands transferred to land poor or landless or needy people, there are more productive income earning opportunities by either tilling lands and engaging in agricultural and related activities (Arnot, Luckert and Boxall, 2011) or using lands as collaterals to establish/enhance farming operations (Feder, 1988; Place, 2009; Deininger and Feder, 2001). These increases the opportunity cost of engaging in protests which would reduce the likelihood of joining such skirmishes.

We use the following specification similar to equation (2) where the outcome variable is replaced as land redistributed in percentage terms taken from Albertus (2015). Given that land redistribution was one of the important land reform motives undertaken in Latin American region countries, it makes sense to pinpoint if the amount of land redistributed could work as a channel in reducing protest intensity.

$$Land\ redist_{it} = \pi_i + \mu_t + \emptyset.\mathrm{tr} + \eta.LR_{it} + \boldsymbol{\theta_2}.\boldsymbol{Z} + \varepsilon_{2it} \quad (3)$$

The proportion of lands redistributed due to land reform implementations would work as a mechanism to mitigate rural protests. We expect the signs of cumulative land reform implementations (η) to be positive from equation (3) above.

The second pathway sheds lights on reducing land inequality which is the highest in Latin American region countries (Binswanger et al. 2009). Here we expect that cumulative land reform implementations would make land distribution more egalitarian and hence the land inequality would diminish. This in turn would reduce the rural protests as the extant literature find a positive association between land inequality and conflict in general (Russett 1964, Verwimp, 2005, Albertus et al. 2018) and particularly in Latin America (Jensen and Sorensen, 2011).

We employ the following specification with land gini coefficient as the outcome variable and the cumulative impact of land reform implementations as the main explanatory variable along with the controls used in the main specification in equation 2.

$$Landgini_{it} = \pi_{1i} + \mu_{1t} + \vartheta_1.\mathrm{tr} + \eta_1.LR_{it} + \boldsymbol{\theta_3}.\boldsymbol{Z} + \varepsilon_{3it} \quad (4)$$

We expect the signs of cumulative land reform implementations (η_1) would be negative as land inequality would decline after such reforms. The data sources details are provided in the following sub-section.

3.5. Data

3.5.1. Dependent variable

The main dependent variable is the large-scale rural protests (denoted by *Lsrp* in equations 1 and 2) taken from Albertus (2015). Albertus (2015) notes that such protests are the major ones, and we take these major protests in the analysis. The protest years are denoted by 1 and the non-protest years are assigned zeros.

In deciphering the mechanisms, two separate dependent variables are employed in equations 3 and 4. These are (i) the amount of land distributed in percentage terms from Albertus (2005) (denoted by *Land redist.* in equation 3) and (ii) the land Gini inequality measures from Frankema (2010) (denoted by *Landgini* in equation 4). The original data for land gini are reported for a few years. We interpolated the intermediate years of data using a linear interpolation method, similar in spirit of Albertus et al. (2018).

In the placebo test, the dependent variable is an alternative indicator of civil strife incidence, which are taken from the Armed Conflict Database of the International Peace Research Institute, Oslo, Norway and the University of Uppsala, Sweden. Miguel et al. (2004), Brückner and Ciconne (2011) and Hull and Imai (2013) employ this incidence of civil conflict measure in their empirical analysis. Following Hull and Imai (2013), we use conflict incidence as an indicator variable taking the value of 1 if there is an ongoing civil conflict in country i at time t which results into at least 25 deaths per year; else, it takes the value 0.

3.5.2. Independent variable

The independent variables are various motives of land reforms implemented from Bhattacharya et al. (2019) and two particular motives of land reforms from Albertus (2015). All land reforms which are partially implemented from Bhattacharya et al. (2019) have been treated as fully implemented in the empirical analysis. The detailed description about the land reform motives has been provided earlier.

3.5.3. Control variables

The novel control variable, democratic participation through Church or clergy-initiated activities (CIDP) is constructed using a number of relevant resources as detailed in the Data Appendix. The years of CIDP or active CIDP are coded as 1, else it takes the value 0.

The other control variables are selected based on the extant literature (Blattman and Miguel, 2010; Esteban et al. 2012). These are: (i) one year lag of log GDP per capita from Maddison (2009) database, (ii) log of population from Maddison (2009) database, (iii) democracy indicator which takes the value of 1 if the relevant polity2 score is equal to or above 5, and 0, otherwise based on Esteban et al. (2012), (iv) oil exports indicator dummy which takes the value of 1 if a country's oil export revenue in that year is at least 1/3 of total export revenue based on Fearon and Laitin (2003), (v) a non-contiguous territory dummy based on Fearon (2003) which is given a value of 1 if any country with a territory of at least 10,000 people and

separated from the mainland containing the capital city either by land or by 100 kilometres of water, else 0 and (vi) the ethno-linguistic fractionalization measure from Fearon (2003).

4. Results and discussion

Figures A to D lay out the results from event study analysis using equation 1. Tables 2 to 11 present the results from empirical analysis along with various robustness checks and placebo test findings. Table 1 contains the summary statistics.

4.1. Results from event study analysis

Figures A (Figure 1 to Figure 6) report the findings with large-scale rural protests as the outcome variable. A quick look at Figure 1 shows that there are no pre-trends, and the post land reform implementations are reducing protests after two years of reforms, especially between years 3 to 7. If we consider the redistributive reform in Figure 2, however, there is no evidence of rural protest mitigation after such land reform implementation within the 10-year time period. This is quite plausible as generally it takes a bit more time (or more than the stipulated 10-year window selected here) to absorb the impact of redistributive reforms. Now, focusing on combined land ceiling, expropriation and redistributive reforms together in Figure 3, again, there is no pre-trend before the land reform implementation window. Similar to Figure 1 above, the post land reforms years show ameliorating effect for rural protests for the lead periods of years 5 to 7.

The most interesting finding is related to the global major/minor land reform by Albertus (2015) in Figure 4. Here we see clear mitigating impact throughout the post 10-year reform implementation period. However, there is some evidence of negative pre-trend in the pre-reform period which raises some concerns on treating the land reform implementations as purely exogenous treatments.

Now focusing on pro-poor land reform as a package in Figure 5, there seems to be muted intervention support as land reform implementations are statistically insignificant for all leads except for years 5 to 7. Note that there is no evidence of pre-trends either in this particular motive of land reform as well. Finally, when we shed light on tenure security improvement reforms in Figure 6, again, there is not statistically significance evidence on mitigating rural protests before or after ten years of such reforms being implemented. Note that tenure security improvement does not entail physical transfer of land, and hence the pacifying impact of such initiatives seems to be muted.

Overall, the event study primer check in Figures A show some evidence of mitigating impact of land reform implementations (in four out of six different motives of land reform implementations analyzed) by focusing on these implementations as exogeneous treatments, especially after 5 to 7 years of such reforms being implemented.

Figures B (Figure 7 to Figure 12) present findings from placebo analysis with conflict incidence as the outcome variable. At the outset, the focus here would be either no impact or maybe positive impact on conflict incidence in the post 10-year window of land reform implementations. Looking at Figure 7 with all land reforms implemented, we find largely statistically insignificant impact in the post reform period with some evidence of positive effect in further leads 6 and 7. Similarly, for redistributive reforms in Figure 8, there is positive impact from periods 6 to 9 in the post 10-year window and the rest of the periods

remain statistically insignificant. The combined motive of land ceiling, expropriation and redistribution in Figure 9 also point to the same narrative as the redistributive land reforms. The global major/minor reforms from Albertus (2015) in Figure 10 show clear support for our placebo test as well with the post reform period's impact on conflict incidence predominantly positive and statistically significant and particularly from the years 3 to 9 in the post 10-year window. In line with the above, the pro-poor land reforms also reveal positive and statistically significant impact in the post reform lead periods of 6 to 9 with no significant impact beforehand. Thus, the placebo checks are in line with the conjecture that land reforms implementations would only ameliorate large-scale rural protests, but not conflict incidents which could be triggered by various reasons apart from rural skirmishes. Unfortunately, all the above reforms are also plagued with statistically significant pre-trends. Thus, even if the post reform findings support the placebo conjecture, we need to be mindful that the reform implementations could not be used as exogenous treatments.

Note that the evidence from tenure security improvement type land reforms in Figure 12 reveals no statistically significant impact both 10-year before and after such reform implementations, and hence lends credence to the falsification test conjecture that land reforms are going to ameliorate only rural protests and no other conflicts.

Figures C (Figure 13 to Figure 18) report the first mechanism primer figures with land redistributed as the dependent variable. There is immediate support from Figure 13 that land reform implementations as a package remains statistically significant and positive in the post reform years without any pre-trends, thus, lending credence to the proposed channel in mitigating rural protests. The redistribution motive land reforms from Figure 14 also show positive support to the proposed mechanism as most of the post reform years reveal positive and statistically significant land redistributions. There are no pre-trends, and hence, redistributed reforms could be termed as exogenous treatments. Similarly, taking a look at the combined land ceiling, expropriation and redistributive motive reforms together in Figure 15, there is some muted support for positive land transfers at leads 2 and 3, but not for the whole 10-year post reform period. Note that there are no pre-trends in the pre-reform implementation time period as well.

In line with the Figure 4 above, the major/minor reforms of Albertus (2015) in Figure 16 reveals clear and sustained improvement in land redistribution in the post 10-year reform period without any evidence of pre-trends. Thus, major/minor reforms of Albertus (2015) could be used as an exogenous treatment to decipher the plausible mechanism.

However, looking at the pro-poor land reforms implementations exclusively, there is no support that such land reform implementations lead to transfers of land even within the 10-year time period though there is clear evidence of no pre-trends in the pre reform implementation period. On the other hand, interestingly, tenure security improvement is now showing positive and statistically significant relationship with land redistribution within the 10-year post reform period without any pre-trends. Hence tenure security improvement could be used as a plausible mechanism.

To summarize, focusing on Figures C, there is very good support that land reform implementations could be used as exogeneous treatments to tease out the potential mechanism as in all cases, there are no pre-trends. Moreover, in five out of six different

motives analyzed in the event study setting, there is positive support for land redistributions in the post reform period.

Figures D (Figure 19 to Figure 24) present the event study primer results for second pathway where land gini coefficient is the outcome variable with the treatment being various land reforms implemented. A quick look at all the land reforms implemented as a package in Figure 19, however, does not support this conjectured channel. Similarly, there is no support from the perspective of redistributive land reform (Figure 20), the combination of land ceiling, expropriation and redistributive reforms taken together (Figure 21), the global major/minor reforms from Albertus (2015) (Figure 22), pro-poor land reforms (Figure 23) as well as the tenure security improvement focused reforms (Figure 24). The no statistical significance could be attributed to the fact that distribution of landholding patterns takes a long time for adjustment, and hence analyzing within the ten years' time interval is a relatively short period for revealing any statistically significant impact.

4.2. Findings from regression models

Table 2 shows the findings from least squares regression which is estimated in the spirit of a two-way generalized difference-in-difference model after controlling for country-specific fixed effects and time effects. The focal variable of interest is the cumulative effect of land reforms implemented with a number of heterogeneous motives as depicted in the rows of the tables.

In Table 2, we report only the heterogeneous land reforms implemented along with the new control variable, the active years of democratic political participation (CIDP) including the years of active CEB.[17] Note that none of the land reforms implemented has statistically significant effect at the conventional level, though the signs are negative for all land reforms, except for the 'Combined end use' reform implementation. Only land reform implemented as a package (LR implemented) has the p-value of 0.13, pointing that the cumulative impact of land reform as a package (with all different motives subsumed) works as a pacifier for large scale rural protests only at 13% level of significance. The statistically insignificant findings could be attributed to possible endogeneity concerns due to omitted variables or reverse causality. Given that we have controlled for a gamut of fixed effects and other relevant variables, including a novel control, CIDP, the possibility of endogeneity arising out of omitted variables would be quite less. Nevertheless, it remains a possibility. In regard to reverse causality, i.e., whether large scale rural protests lead to land reform implemented rather than the other way around, there is no evidence of such incidents from the original data source (Albertus 2015). There could be some drives towards land reforms legislations being passed or enacted due to protests, however, the chance of implementing such reforms in a cumulative way would be a big ask.

The new control variable, years of active clergy induced democratic participation (CIDP) coefficients reported in the table show one interesting fact: that years or active years of democratic participatory action on the ground, do indeed dampen large scale rural protests, albeit not in a statistically significant way except for the 'Combined end use' type land reforms (specification 10, Table 2). The CIDP estimate is very close to the 10% significance level when used as a control with land reforms implemented as a package (LR implemented),

[17] The full set of results for all control variables are reported in the Appendix Tables.

redistributive land reforms (Redistributive), global national land reforms (LR national), which again incorporates expropriation and redistribution without looking at the extent of land being transacted) as well as in the land reforms implemented to improve tenure security (Tenure sec. improve).

Next, in Table 3, we present the results from Poisson model with conditional fixed effects to check the robustness of the least squares findings. The results here are better than the least squares in two dimensions. First, unlike in Table 2, now three land reform implementations show statistically significant and negative effect. These are (i) global major/minor reforms, (ii) global national reform and (iii) the tenure security improvement type reforms with the most significant impact attributed to the global major/minor reforms at 99% level of significance. Interestingly, the combined end use land reform implementations are now associated with higher likelihood of protests, which is in deference to our conjecture. Second, the new control variable, CIDPs are consistently negative and statistically significant at 90% level in all specifications except for the global major/minor reform's case. Thus, there is very good support from the Poisson model estimation that large-scale rural protests could be mitigated with particular types of land reforms where either land are physically transferred through expropriation and redistribution (the focus of global major/minor reforms and global national reforms) or through improvement in land tilling arrangement rights by enhancing tenure security. In addition, church induced democratic participation (CIDP) also works as important and direct pacifiers for rural protests.

4.3. Instrumental variable estimation results

The instrumental variable (IV) estimation addresses endogeneity concerns where years of elite split is employed as an instrument for cumulative land reform implementations. Albertus (2015) reports that elite split is a strong predictor of land reforms in Latin America. To be specific, reforms where lands were physically transferred from landowners to landless, are particularly attributed to elite split between landed elites and political elites. The first stage of IV results is reported in Table 4 and the second stage of IV results are presented in Table 5.

4.3.1. First stage results

Taking a look at Table 4, we find that elite split is a strong instrument for a number of heterogeneous land reform implementations except for the two, the land ceiling and the combined end use type reforms. The KP Wald F-statistic and the Cragg-Donald Wald F-statistic both show that elite split is a relevant instrument given the F-statistics are much above the Stock-Yogo critical values. In terms of magnitude, the elite spilt coefficient has the highest value for the global major/minor reform (LR maj-min.) reform which is a land reform categorized in Albertus (2015). This makes sense as a large proportion of land reforms implemented in Latin America saw lands being expropriated and redistributed. Note that, even individually, the expropriation and redistributive land reforms from Bhattacharya et al (2019) in specifications 3 and 4 are also showing 99% statistical significance.

In the same vein, if we look at the combined motive of land ceiling, expropriation and redistribution in specification 5, we find that elite split is also an important predictor of such reform combination. Ignoring the major/minor aspect of land reform of Albertus (2015) in specification 6, if we concentrate only on land redistribution of Albertus (2015) in specification 7 (LR national, which again incorporates expropriation and redistribution

without looking at the extent of land being transacted), there is also strong evidence that elite split is working as an appropriate instrument.

Land reforms listed in specification 3 to specification 7 involve physical transfer of land to the beneficiaries. In specifications 1 and 8, land reform implemented as a package subsuming all motives (LR implemented) and land reform targeted to the poor (Propoor), respectively, include both physical transfer of land as well as improving land users rights (like improving tenure security arrangements or legalizing informal land tenure arrangements) without reassigning any physical land. The instrumental variable, elite split, is strongly correlated with such reforms as well. Similarly, the tenure security improvement motive could also be instrumented with elite split, though it did not pass one of the first stage F-tests, KP Wald rk F-stat. Overall, it is heartening to know that elite split works as a strong instrument for a number of different motives of land reforms, be it involving physical transfer of land or transfer of some users rights, as well as combination of both types of categories.

One plausible reason regarding the two land reforms which did not pass the first stage diagnostics (land ceiling and combined end use land reforms) would be that these reforms were implemented sparsely in Latin America. Lipton (2009) and Bhattacharya et al (2019) mention that land ceiling motive land reforms were prevalent in South Asian countries, and not in the Latin American countries. The combined end use land reforms are the type of reforms where end user rights to lands were transferred to the beneficiaries. These are the most progressive types of land reform in the sense that the beneficiaries not only have physical possession of land, but also enjoy total control of landownership and they are free to use these lands in any way they want. Based on Bhattacharya et al. (2019), these types of land reforms are land consolidation, land distribution, land privatization and land restitution. Again, these types of reforms were hardly implemented in the data sample.

The results from the new control variable, CIDP, are also presented in Table 4. The idea here is that the active democratic participatory organizations including CEBs would work as facilitators for land reform implementation activities. The CEBs undertook a number of different political undertakings apart from offering prayers such as 'human rights monitoring, organizing labor protests, establishing cooperatives and credit unions, providing health care and education, fundraising, supporting solidarity campaigns, publishing and other reform activities' (Bruneau, 1980; CIA Report 1986). Thus, years of CIDPs could be seen as an impetus to big ticket decisions like land reforms. In line with the above conjecture, we find that for global major/minimum reforms (LR maj-min.), global national reforms (LR national) and expropriation motive reforms, CIDP plays statistically significant and positive roles. The CIDP also shows positive signs for a number of other reforms like Redistributive and Propoor but remains statistically insignificant. Given the nature of reforms involving physical transfer of land (expropriation and redistribution), it appears that year of active CEBs played important roles in implementing such decisions. Bruneau (1980) mentions how CEBs in Brazil and other Central American countries are driving the local development agenda including establishment of production and consumer cooperatives and hiring lawyers to challenge arbitrary political and administrative measures. The CIA document (1986) highlights active role plays by CEBs in (1) Nicaragua in the 1970s "*to promote social action and apply pressure to Somoza government*", (2) El Salvador in the 1980s to give support to democratic reform initiatives, and (3) Guatemala in the 1970s in the wake of widespread human rights abuse by the successive oppressive military regimes. Note that for El Salvador

the large scale land reforms were not implemented (Kay, 2001) unlike in other countries like Mexico, hence CIDPs could instigate such decisions.

4.3.2. Second stage results

The second stage results from the IV regression are presented in Table 5. The outcome variable is the years of large-scale rural protests (denoted by *Lsrp*). All ten different land reform implementation coefficients are reported. Note that first stage results are passed for all models expect for model 2 (land ceiling land reform motive) and model 10 (combined end use motive land reform). There is partial support for first stage in model 9 involving tenure security improvement type land reforms as the Cragg-Donald Wald F-statistic value is more than the threshold level but the KP Wald rk F-statistic value is less than the threshold level.

Looking at the estimated coefficients, there is overwhelming support for cumulative land reform implementations to be used as protest mitigators as in all cases, except for the combined end use motive, the land reform implementations are showing negative and statistically significant effect on rural protests. The land reforms used as a package subsuming all different motives of reforms (LR implemented) in model 1 reveal that a standard deviation increase in the cumulative land reform implementation reduces demonstrations by 0.911, i.e., 0.81 standard deviations of the large-scale rural protests. Now looking at pro-poor land reforms (Propoor) from model 8, the magnitude comes to reduction of 0.86 standard deviations of the large-scale rural protests due to a standard deviation increase in the cumulative pro-poor land reform implementation. The magnitudes are relatively large. The findings here are in line with the earlier short-term conjecture from the event study analysis in Figures A. Thus, it is interesting to note that within a short time horizon in the event study setting and using a significantly large time window in the endogeneity corrected IV analysis, there is very good support that land reforms could be used to mitigate low-intensity conflicts like the large-scale rural protests. This resonates with the studies like Albertus and Kaplan (2003) for Colombia and Albertus (2020) for Peru.

Now focusing on land reforms where lands were physically transferred through redistribution (Redistributive, model 4), it reduces large-scale rural protests by 0.67 of standard deviation. Similarly, one standard deviation increase in the cumulative expropriation type land reform implementation in model 3 diminishes protests by around 0.90 of standard deviation. Taking landceiling, expropriation and redistributive motives together in model 5, it leads to a reduction of protests by 0.88 of standard deviation. Hence, land reforms where lands are physically transferred lead to even higher mitigation of large-scale protests than the reforms subsuming different motives or pro-poor reforms.

Taking a look at the global major/minor reforms in model 6, there is 0.19 of standard deviation decline in the large-scale rural protests. Similarly, the one standard deviation increase in the global national reform implementation leads to 0.24 of standard deviation mitigation of rural protests. Note that these two types of motives incorporate expropriation and redistribution type reforms together with a threshold level of land being transferred. In such instances also, we find support that land reform implementations mitigate rural protests effectively.

The tenure security improvement reforms enhance tenants' engagement in the land tilling arrangements with landowners without transferring physical lands or full end users' rights in

lands as owners. Yet, in such instances also, the rural protest is dampened by the highest extent, 2.51 of standard deviation. This is an interesting finding as the event study results show no statistically significant impact within the 10-year before or after window from the event study setup. Hence, it takes a longer time to realize its dampening impact on the large-scale rural protests.

The clergy induced democratic participation (CIDP) including active years of CEB also dampens large-scale rural protests by 0.03 of standard deviation when used as a control in model 1 with all motives of land reforms implemented (LR implemented). The finding reinforces our conjecture that church or clergy induced democratic participation ameliorates protests directly when used as a control in the second stage of instrumental variable analysis. The rest of the CIDP coefficients from other models (model 2 to model 10), however, are not statistically significant though they remain always negative in terms of their signs.

A quick look at the other controls in the appendix Table 3A reveals that one lag of real GDP per capita (GDPl1) also works as a dampener in reducing protests. This is in line with a large number of papers in the extant literature on conflict (Blattman and Miguel, 2010; Verwimp et al. 2009) which establish that income is one of the most important mitigation variables for civil strife. The other interesting finding is involving oil exporting dummy (Oil) which shows that more export revenue from oil is associated with a decline in rural protest incidences except for model 2 (landceiling motive) and model 10 (combined end use motive) which incidentally did not pass the first stage restrictions. Also note that democracies (Democracy) are associated with more rural protests, i.e., democratic regimes enhance rural demonstrations in Latin American countries. However, the magnitudes of such democratic regime effects are small.

4.4. Findings from the placebo test

Tables 6 and 7 lay out the results from placebo test with conflict incidence as the alternative outcome variable. Table 6 provides the first stage results and Table 6 reports the second stage results. These falsification evidence will bolster the exogeneity property of the instruments in the sense that we expect that the instruments (elite split) used for land reform implementations would only pacify the rural protests in the second stage, but these would not mitigate conflict incidences in general and may even aggravate such conflict incidences at the second stage when instrumented for land reform implementations.

The first stage results from Table 6 reinforces the veracity of elite splits as strong and relevant instruments for land reform implementations in all models except for land ceiling and combined end use motive reforms (note that these are not reported in Table 6). The elite splits are strong and relevant instruments in all different motives of land reforms with the first stage estimates are statistically significant at 99% and all the relevant first stage test statistics are passed comfortably. The findings are in line with first stage main results from Table 4 earlier.

Interestingly, the novel control variable, CIDP also shows positive and statistically significant effect for expropriation and global major/minor reforms, similar to main findings from Table 4. Note that in contrast to Table 4 results, now in case of global national land reform, the CIDP is not statistically significant though the sign becomes positive.

The second stage results from Table 7 reveal that various land reform implementations are instigating more conflict incidents when instrumented with elite splits. The findings immediately support the falsification exercise that cumulative land reforms implementations would mitigate only large-scale rural protests, but not conflict incidents. The second stage results are in line with de Janvry et al. (2015) for Mexico and Albertus and Kaplan (2013) for Colombia.

Reconciling the magnitudes from Table 7 with the main findings from Table 5, we find that for cumulative land reform implementation as a package (LR implemented) instigate conflict incidence by 0.66 of standard deviation. Note that from the main results table (Table 5), one standard deviation increase in LR implemented leads to rural protests reduction by 0.81 of standard deviation. Hence, cumulative land reforms implemented have a net reduction effect of 0.15 of standard deviation. Thus, the falsification or placebo conjecture is established in terms of the magnitudes as well. Similarly, considering pro-poor land reforms as a package (Propoor), there is 0.61 of standard deviation increase in the conflict incidence, however, from the main findings in Table 5, the cumulative pro-poor land reform implementations are associated with 0.86 of standard deviation mitigation in rural protests. Hence, in this instance also, there is overwhelming evidence of net reduction in rural protests which bolsters the placebo evidence.

In line with the above, we find net reduction in rural protests (reported in parentheses) for all other land reform implementations involving physical transfers of land, viz., expropriation (0.37 of standard deviation), redistributive (0.25 of standard deviation) as well as for combination of land ceiling, expropriation and redistribution (0.26 of standard deviation). Interestingly, for two other reforms where lands were physically transferred as well, viz., the global major/minor reforms and the global national reforms, the net reduction in magnitudes is closer to zero, i.e., the effect of such reforms are reducing the large-scale rural protests and instigating the conflict incidences at the same extent.

Looking at the tenure security improvement reforms, there is also net reduction in rural protests evidence of 1.04 of standard deviation due to such cumulative reforms. This is a very interesting finding as in such instances, there is enhancement of land tilling arrangements without transferring end-user rights in land use. Also note that the clergy induced democratic participation (CIDP) is never statistically significant and remains mostly positive in Table 7. This provides some support to our placebo analysis as well in the sense that this novel instrument is only mitigating rural protests in the second stage main analysis (Table 5, all coefficients negative) and does not affect the conflict incidence at all.

5. Mechanisms

5.1. Mechanism I: Improvement in percentage of land redistributed

Table 8 and Table 9 provide results from first stage analysis and second stage estimation using equation 3 with percentage of land redistributed as the dependent variable and cumulative land reforms implemented as the main explanatory variables. Given the main findings that cumulative land reforms do dampen large scale rural protests, it is natural to figure out a plausible channel or mechanism on how such mitigating impact could be transpired through.

Taking a look at the first stage results (Table 8), there is overwhelming support that elite splits are very effective, strong and relevant instruments for cumulative land reforms implementations, a consistent finding reported throughout the empirical analysis in the paper. Also, the clergy induced democratic participation (CIDP), the novel control variable, remain positive and statistically significant predictor of such reforms for two motives, expropriation and global major/minor types of initiatives. For all other motives barring LR implemented, the CIDP variable remains positive though not statistically significant.

Table 9 estimation results provide unequivocal support to percentage of land redistributed due to cumulative land reforms implementations as a credible mechanism. In all eight different models involving a variety of reforms motives, we find that reforms lead to positive and statistically significant land transfers. We can interpret these transfers in terms of extensive margins, i.e., lands redistributed provided additional opportunities for land tillers in engaging productive income earning opportunities which subsequently mitigated widespread rural protests.

In regard to magnitudes, one standard deviation increase in the cumulative land reform implemented as a package (LR implemented, model 1) leads to 0.09 of standard deviation increase in land transferred. Given that the standard deviation of land redistribution is quite high (3.5827, Table 1), the above magnitude could be interpreted as 0.32 standard deviation increase in land redistributed which is a reasonable amount of land being transferred as a result of such cumulative reforms. Similarly, a standard deviation increase in the cumulative pro-poor land reform implementations as a package (Propoor, model 7) lead to a 0.37 standard deviation increase in land redistribution. Note that both of these magnitudes of the land reforms explain around half of mitigating effects (0.81 and 0.86 of standard deviations) from the main analysis in Table 5. Thus, lands transferred as packages remain very effective as mechanisms in reducing large-scale rural protests.

Now focusing on other motives involving physical transfers of lands, there is 0.41 of standard deviation increase in land transfers as a result of expropriation (model 2), 0.42 of standard deviation increase in land transfers due to redistribution (model 3), 0.37 of standard deviation improvement in land transfers due to combined land ceiling, expropriation and redistribution motive land reforms (model 4), 0.05 of standard deviation increase in land transfers due to global major/minor reforms (model 5) and 0.08 of standard deviation improvement in land transfers due to global national land reforms (model 6). Amongst these motives, it is not surprising that redistributive motive is associated with the highest magnitude of land transfers as these transfers were largely attributed with redistribution. Taken together, there is overwhelming support for land redistribution as a channel as all of the above land reforms are reducing large-scale rural protests in the main analysis highlighted before (Table 5).[18]

The above findings in terms of land redistributions have traction from the extant literature (Binswanger et al. 2009; Kay, 2001). Kay (2001) highlights the heterogeneities in terms of land transfer in Latin America with countries like Mexico, Guatemala, Chile, Peru and

[18] Note that the tenure security improvement land reform is also leading to land transfers. However, given the first stage coefficients of such reforms are not of very high magnitude, we can safely ignore the magnitude involving such transfers.

Guatemala have embarked on large-scale land redistributions whereas countries like El Salvador and Colombia did not transfer that much of land.

Table 9 findings also highlight the role played by the novel control variable, the church or clergy induced democratic participation (CIDP) in regard to such land transfers. In all different motives of reforms, except for expropriation, there is positive and statistically significant effect of CIDP to land redistributions. The effect is the strongest when used with cumulative land reforms implemented as a package (LR implemented, model 1) with one standard deviation increase in CIDP causing around 0.10 of standard deviation increase in the natural log of land redistributions in percentages. This is followed by usage in conjunction with the combination of land ceiling, expropriation and redistributive reform (Ceiling, expro. & redist, model 4) with one standard deviation increase in CIDP leading to around 0.08 of standard deviation increase in natural log of land transfers in percentages. The CIDP control also remains influential when employed with redistributive (model 3) and propoor land reforms (model 7) with the magnitudes of 0.07 of standard deviation increase in natural log of land transfers in percentages and 0.06 of standard deviation improvement in natural log of land redistributions in percentages, respectively.

In contrast, the CIDP variable, when used with global major/minor reforms (model 5) and global national land reforms (model 6), show a bit lesser impact in terms of land redistribution percentages (around 0.04 of standard deviation increase and around 0.06 of standard deviation, respectively), but the effect remains positive and significant. Note that church induced democratic participation involved multifaceted activities in various dimensions like setting up production and consumer cooperatives, hiring lawyers to challenge arbitrary political and administrative decisions, fighting against human rights abuses as well as lobbying for democratic reforms (Jordan, 2022). The CIDP results here show one such manifestation through transfer of land which is quite remarkable.

5.2. Mechanism II: Reduction in landgini coefficient

The second mechanism sheds light on reduction in land distribution through cumulative implementations of such major land reforms. The intuition is that with significant reform implementations which entails transfers of land, the land distribution would improve over time, thus reducing the land inequality manifested by the land gini coefficients. Albertus, Brambor and Ceneviva (2018) present latest evidence on land inequality in a number of Latin American countries and find that the relationship between land inequality and rural unrest in Brazil is quite unique in the sense that one-off land invasions are positively linked with land inequality; however, sustained land invasion threats could be mitigated by landowners, thus showing a negative relationship between rural skirmishes and land inequality.

Our channel idea is rooted with the later evidence from Albertus et al (2018) as we expect that major land reforms would reduce landholding inequality and this in turn would reduce the likelihood of large-scale rural protests. This is in spirit with the famous ‘greed vs. grievance’ hypothesis put forward by Collier and Hoeffler (2004). The first stage results are presented in Table 10 and in Table 11, we lay out the second stage results.

The first stage results from Table 10 again supports the elite splits employed as instruments for various land reform implementations. The effect of the split is the highest for global major/minor reform followed by the global national land reform. The magnitude for land

reform as a package, pro-poor land reform as a package and expropriation reforms remains very similar when instrumented with elite split. The lowest magnitude is attributed to tenure security improvement type reforms.

The church or clergy induced democratic participation (CIDP) remains positive and statistically significant for global major/minor reforms and expropriation type reforms. This finding is in line with first stage mechanism results reported in Table 8. Thus, CIDP matters in pushing through such land reforms implementations. Note that the coefficients for CIDPs remain positive for all other reform implementations though these are not statistically significant.

The second stage results from Table 11 reveal immediate support to our conjectured mechanism. In line with a priori expectation from equation 4, we find that the coefficients associated with various land reform implementations are always negative and statistically significant. Hence, these cumulative land reforms implemented reduced land inequality in a statistically significant way. Reconciling the first and second mechanism results, we can interpret the findings from Table 11 as 'intensive margin' results akin to the trade literature in the sense that not only lands were transferred (as per the first mechanism, could be termed as extensive margin), but such transfers actually made the land distributions more egalitarian.

Focusing on the magnitudes, a one standard deviation increase in the cumulative land reforms implemented as a package (LR implemented, model 1) is reducing the land gini by 0.005 of standard deviation. This magnitude remains exactly the same when we look at the pro-poor land reforms as a package in model 7. Though the estimates seem to be small, but from the context of Latin America where the land gini coefficient is quite high (the average for our data sample is 0.76), even these small changes would have bigger impacts in the long run. The findings here support earlier conjecture of Wood (2003) that land reforms would work as a deterrent to rural protests by reducing land inequality. Note that Albertus and Kaplan (2013) and Albertus (2020) investigate the heterogeneities between small-scale and large-scale land reforms and civil conflict in Colombia and Peru, respectively and highlight that only large-scale land reforms which were implemented had mitigating effect on civil strife. Given that these major reforms implementations were marred with some political interests, the subsequent reductions in land gini coefficients could not be that much as our results depict.

Taking a look at the reforms where lands are transferred physically, we find that one standard deviation increase in redistributive reforms in model 3 leads to around 0.004 of standard deviation reduction in land gini coefficient. Similarly, expropriation type reforms cumulative implementation (model 2) leads to around 0.004 of standard deviation reduction in the land gini coefficient. The combination of land ceiling, expropriation and redistribution reforms in model 4 also reduces land gini coefficient by 0.005 of standard deviation. The two other reforms entailing transfers of land, viz., global major/minor reform (model 5) and global national reform (model 6), also diminishes the land gini by the same extent, 0.002 of standard deviation. There is a bit of heterogeneity in terms of reduction in land gini coefficient, but this is expected given the diversified nature of the reforms pursued. Interestingly, tenure security improvement motive reforms are also reducing land gini coefficient by 0.01 of standard deviation which shows that enhancing land tilling arrangements bode well for

landholding distributions. The findings resonate with conjectures from Huntington (1968) and Kalyvas (2006).

Note that clergy or church induced democratic participation (CIDP), the novel control variable, is not statistically significant in any of the eight models used. In addition, their magnitudes are closer to 0. This is quite plausible as CIDPs open up some democratic windows of opportunities for supporting policies such as land reforms implementations as was manifested in the first stage results (Table 10); however, such involvement could not be directly transpired to changing land distributions and hence affecting the land gini coefficients subsequently.

6. Conclusion

The study analyses land reforms as potential pacifiers of large-scale rural protests in Latin America. The rural protests were quite prevalent and could be termed as low-intensity civil strife given that human casualties were largely absent. We investigate if major land reforms which were implemented and their cumulative impact in mitigating such skirmishes in the region over a long period of time. Exploiting a gamut of heterogenous land reforms from Bhattacharya et al. (2009) and Albertus (2015) and controlling for a number of potential confounders including time-varying controls and a host of fixed effects as well as time trends, we find robust evidence that cumulative land reforms implemented have mitigating effect on large-scale rural protests. The findings are in line with Albertus (2020) and Albertus and Kaplan (2013).

The empirical strategy sheds lights on both short-run effects as well as long-run implications. The short-run effects are revealed with a flexible event study analysis with land reforms implemented as treated and no reforms implemented as control groups within a 10-year before and after window. Even within this short time span, we find some support that land reforms remain effective as a protest mitigation mechanism. The full scale long-run analysis using two-stage least squares employ years of elite spilt from Albertus (2015) as exogenous instruments for land reforms implemented.

Amongst the land reforms, the strongest impact is attributed to the redistributive and expropriating type reforms individually as well as the combination of such reforms implemented jointly. In addition, we find that tenure security improvement type reforms which do not entail physical transfer of land and focus on enhancing tenure arrangements between land tillers and landowners also has a detrimental impact on large-scale rural protests.

We propose a novel control variable, the church or clergy induced democratic participation (CIDP) including the years or active years of base Christian communities which provides a positive impetus to land reform implementations in our first stage instrumental variable analysis. Consequently, the coefficients of CIDPs become negative in the second stage instrumental variable regressions, lending credence to the idea that active years of democratic participation have a direct and soothing impact on large-scale rural protests. In the paper's context, the CIDP heralds in a new democratic window of opportunity. This evidence goes well with the fact that Church participation plays an important role in shaping socio-economic and political decision making in Latin America.

In regard to mechanisms, we pinpoint two plausible pathways involving proportion of land redistributed from Albertus (2015) and reduction in land inequality using data from Frankema (2010). First, there is strong evidence that land reforms implemented redistributed lands to the needy, thus creating more productive income earning opportunities (akin to the extensive margin type argument from international trade), which in turn reduced rural protests. Similarly, the land inequality gini coefficients show statistically significant reductions in land holding distributions, akin to an improvement in intensive margin of land holding patterns. In line with the main results, redistributive and expropriation type reforms as well as their combinations are associated with both higher proportions of land being redistributed as well as more reductions in land inequalities. Thus, these motives are attributed to higher protest mitigations in the main analysis.

This paper's investigation complements the extant literature on land reform and civil strife in Latin America by analysing the cumulative impact of major land reform implementations as a plausible tool to mitigate low intensity conflict manifested by large-scale rural protests. Given the huge debate surrounding the efficacy of land reforms to ameliorate rural skirmishes in this very important region of the world, the study sheds lights on possible pathways through which small scale protests could be mitigated.

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Tables

Table 1. Descriptive statistics

VARIABLES	Mean	Standard deviation	Minimum	Maximum	Sample size
Dependent variables					
Lsrp	0.1777	0.3823	0	1	2,004
Conf.	0.1487	0.3560	0	1	968
Land redist.	0.6961	3.5827	0	62.8	1,467
Landgini	0.7629	0.0946	0.291	0.865	1,207
Independent variables					
LR implemented	0.5837	0.8899	0	4.5	2,204
Land ceiling	0.1284	0.3182	0	1.5	2,204
Expropriation	0.3886	0.7479	0	3.5	2.204
Redistributive	0.3140	0.5125	0	2	2,204
Ceiling, expro. & redist.	0.4736	0.8288	0	4.5	2,204
LR maj-min.	0.2759	0.5889	0	2	1,975
LR national	0.2025	0.4019	0	1	1,975
Propoor	0.4745	0.8344	0	4.5	2,204
Tenure sec. improve	0.2291	0.6817	0	3.5	2,204
Combined end use	0.1733	0.4222	0	1.5	2,204
Instrumental variable					
Elite split	0.3348	0.4720	0	1	2,109
Control variables					
CIDP	0.1095	0.3122	0	1	2,220
GDPl1	7.8829	0.5711	6.3315	9.2111	1,164
Lpop	8.9767	1.2140	6.5439	12.4221	1,254
Oil	0.1164	0.3207	0	1	1,186
Nonconti	0.0535	0.2251	0	1	1,234
Fractionalization	0.4292	0.1814	0.1322	0.7434	1,186
Democracy	0.4573	0.4983	0	1	1,231

Note: 'Lsrp' stands for years of large-scale rural protests from Albertus (2015). It takes the value of 1 during year/years of protests and 0 otherwise. 'Land redist.' denotes percentage of land redistributed from Albertus (2015). 'Landgini' is the land gini values from Frankema (2010) which are interpolated for intermediate years as in Albertus et al. (2018). 'Conf' refers to conflict incidence variable as in Hull and Imai (2013). 'LR implemented' denotes land reforms implemented with various motives incorporated as a package; 'Land ceiling' represents land reforms implemented with land ceiling motives; 'Expropriation' denotes land reforms implemented with expropriation motive; 'Redistributive' denotes land reforms implemented with redistributive motive; 'Ceiling, expro. & redist.' denotes the land reforms implemented with the combined motives of land ceiling, expropriation and redistribution; 'LR maj-min.' denotes global major/minor land reform from Albertus (2015); 'LR national' depicts global national land reform from Albertus (2015); 'Propoor' depicts land reforms implemented with pro poor motive; 'Tenure sec. improve' denotes land reforms implemented with tenure security improvement motive; 'Combined end use' denotes land reforms implemented where end users rights were transferred to beneficiaries; 'Elite split' denotes years of elite split from Albertus (2015); 'CIDP' denotes the active year/years of the clergy initiated democratic participation initiatives including CEBs; 'GDPl1' is the one lag value of log per capita GDP; 'Lpop' is log of population; 'Oil' is a dummy variable which takes the value 1 if a country earns 1/3 of its exports revenue from oil in that particular year; 'Nonconti' stands for non-contiguous territory dummy in a country taking the value of 1 if there is a non-contiguous territory and 0, otherwise; 'Fractionalization' is ethno-linguistic fractionalization measure taken from Fearon (2003); 'Democracy' depicts democracy dummy which takes the value 1 if the corresponding polity2 score takes the value of 5 or above, else it is coded as 0 in line with Esteban et al. (2012).

Table 2. OLS results

VARIABLES	(1) Lsrp	(2) Lsrp	(3) Lsrp	(4) Lsrp	(5) Lsrp	(6) Lsrp	(7) Lsrp	(8) Lsrp	(9) Lsrp	(10) Lsrp
LR implemented	-0.219									
	(0.139)									
	0.132									
Landceiling		-0.271								
		(0.281)								
		0.348								
Expropriation			-0.122							
			(0.0789)							
			0.140							
Redistributive				-0.084						
				(0.151)						
				0.586						
Ceiling, expro. & redist.					-0.142					
					(0.146)					
					0.346					
LR maj-min.						-0.050				
						(0.0441)				
						0.274				
LR national							-0.012			
							(0.0673)			
							0.862			
Propoor								-0.162		
								(0.145)		
								0.277		
Tenure sec. improve									-0.124	
									(0.211)	
									0.563	
Combined end use										0.031
										(0.130)
										0.812
CIDP	-0.107	-0.101	-0.010	-0.107	-0.104	-0.097	-0.109	-0.101	-0.108	-0.110*
	(0.065)	(0.065)	(0.065)	(0.065)	(0.066)	(0.066)	(0.065)	(0.066)	(0.064)	(0.064)
	0.117	0.141	0.144	0.118	0.129	0.162	0.114	0.141	0.107	0.099
Observations	1,225	1,225	1,225	1,225	1,225	1,206	1,206	1,225	1,225	1,225
R-squared	0.468	0.461	0.461	0.458	0.461	0.486	0.484	0.462	0.457	0.456
Countries	19	19	19	19	19	19	19	19	19	19
Other controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CFE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CTE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CTT	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

Note: 'Lsrp' stands for years of large-scale rural protests from Albertus (2015). It takes the value of 1 during year/years of protests and 0 otherwise. 'LR implemented' denotes all kinds of land reforms implemented as a package; 'LR maj-min.' denotes global major/minor land reform from Albertus (2015). 'LR national' depicts global national land reform from Albertus (2015). Only the new control variable, CIDP is reported for expositional purpose. 'CIDP' denotes the active year/years of the clergy initiated democratic participation initiatives including CEBs. The other controls are, (i) 'GDPl1' is one year lagged log per capita GDP; (ii) 'Lpop' is log of population; (iii) 'Oil' is a dummy variable which takes the value 1 if a country earns 1/3 of its exports revenue from oil in that particular year, (iv) 'Democracy' which takes the value 1 if the polity2 score is 5 or above, else 0; (v) 'Nonconti.' which takes the value 1 if the country has non-contiguous territory and (vi) 'Fractionalization' measure from Fearon (2003) paper. We are not reporting estimated coefficients for all other controls and constant in this table (refer to Appendix Table 2 for full results). Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table 3. Poisson model estimation with fixed effects results

VARIABLES	(1) Lsrp	(2) Lsrp	(3) Lsrp	(4) Lsrp	(5) Lsrp	(6) Lsrp	(7) Lsrp	(8) Lsrp	(9) Lsrp	(10) Lsrp
LR implemented	-0.0529									
	(0.398)									
	0.894									
Landceiling		0.521								
		(0.746)								
		0.485								
Expropriation			0.177							
			(0.385)							
			0.647							
Redistributive				0.229						
				(0.405)						
				0.572						
Ceiling, expro. & redist.					0.103					
					(0.395)					
					0.793					
LR maj-min.						-0.552***				
						(0.212)				
						0.00919				
LR national							-0.568**			
							(0.285)			
							0.0462			
Propoor								-0.0448		
								(0.385)		
								0.907		
Tenure sec. improve									-0.829**	
									(0.349)	
									0.0175	
Combined end use										0.872**
										(0.435)
										0.0452
CIDP	-0.419*	-0.427*	-0.421*	-0.435*	-0.427*	-0.341	-0.412*	-0.419*	-0.453*	-0.392*
	(0.246)	(0.242)	(0.238)	(0.248)	(0.249)	(0.227)	(0.246)	(0.247)	(0.245)	(0.238)
	0.0887	0.078	0.0775	0.0795	0.0861	0.134	0.0935	0.0899	0.0641	0.0999
Observations	1,159	1,159	1,159	1,159	1,159	1,141	1,141	1,159	1,159	1,159
Countries	18	18	18	18	18	18	18	18	18	18
Other controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CFE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CTE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CTT	N	N	N	N	N	N	N	N	N	N

Note: 'Lsrp' stands for years of large-scale rural protests from Albertus (2015). It takes the value of 1 during year/years of protests and 0 otherwise. 'LR implemented' denotes all kinds of land reforms implemented as a package; 'LR maj-min.' denotes global major/minor land reform from Albertus (2015). 'LR national' depicts global national land reform from Albertus (2015). Only the new control variable, CIDP is reported for expositional purpose. 'CIDP' denotes the active year/years of the clergy initiated democratic participation initiatives including CEBs. The unreported controls are, (i) 'GDPl1' is the log per capita GDP; (ii) 'Lpop' is log of population; (iii) 'Oil' is a dummy variable which takes the value 1 if a country earns 1/3 of its exports revenue from oil in that particular year, (iv) 'Democracy' which takes the value 1 if the polity2 score is 5 or above, else 0; (v) 'Nonconti.' which takes the value 1 if the country has non-contiguous territory and (vi) 'Fractionalization' measure from Fearon (2003) paper. We are not reporting estimated coefficients for all other controls and constant in this table (refer to Appendix Table 2ALA for full results). Robust standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1

Table 4. First stage results, baseline

VARIABLES	(1) LR implemented	(2) Land ceiling	(3) Expropriation	(4) Redistributive	(5) Ceiling, expro. & redist.	(6) LR maj-min.	(7) LR national	(8) Propoor	(9) Tenure sec. improve	(10) Combined end use
Elite split	0.140***	0.029**	0.106***	0.097***	0.120***	0.354**	0.187***	0.125***	0.034***	0.006
	(0.021)	(0.011)	(0.025)	(0.020)	(0.020)	(0.048)	(0.030)	(0.021)	(0.009)	(0.013)
	0.000	0.012	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.627
CIDP	-0.011	0.029**	0.063**	0.020	0.018	0.199***	0.066*	0.030	0.005	0.008
	(0.023)	(0.013)	(0.023)	(0.022)	(0.023)	(0.054)	(0.036)	(0.022)	(0.011)	(0.013)
	0.656	0.037	0.006	0.381	0.431	0.000	0.070	0.182	0.626	0.531
KP rk LM stat	42.049	6.879	19.524	22.943	33.127	54.088	37.370	35.172	14.382	0.258
[p-val]	[0.000]	[0.009]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.612]
KP Wald rk F stat	42.225	6.386	18.977	22.283	32.864	54.563	36.517	35.114	13.562	0.236
Cragg-Donald Wald F statistic	66.783	9.649	31.224	35.777	52.921	87.032	51.544	57.990	20.926	0.374
Observations	1,225	1,225	1,225	1,225	1,225	1,206	1,206	1,225	1,225	1,225
Countries	19	19	19	19	19	19	19	19	19	19
Other controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CFE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CTE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CTT	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

Note: The dependent variables are various motives of land reforms implemented. The instrumental variable is the years of elite split from Albertus (2015). Only the new control variable, CIDP is reported for expositional purpose. 'CIDP' denotes the active year/years of the clergy initiated democratic participation initiatives including CEBs; KP stands for Kleibergen Paap tests. The Stock-Yogo critical value for 5% relative bias is 13.91 and 10% relative bias is 9.08. Hansen's J-statistic is not reported as the equation is exactly identified with the only instrument (elite split). CFE represents country-specific fixed effects, CTE represents time fixed effects, CTT stands for country-specific time trend. Robust standard errors are reported in parentheses; *** p<0.01, ** p<0.05, * p<0.

Table 5. IV results, second stage, baseline

VARIABLES	(1) Lsrp	(2) Lsrp	(3) Lsrp	(4) Lsrp	(5) Lsrp	(6) Lsrp	(7) Lsrp	(8) Lsrp	(9) Lsrp	(10) Lsrp
LR implemented	-0.911***									
	(0.209)									
	0.000									
Landceiling		-4.390***								
		(1.692)								
		0.009								
Expropriation			-1.200***							
			(0.349)							
			0.001							
Redistributive				-1.309***						
				(0.356)						
				0.0002						
Ceiling, expro. & redist.					-1.067***					
					(0.262)					
					0.001					
LR maj-min.						-0.317***				
						(0.0838)				
						0.001				
LR national							-0.599***			
							(0.170)			
							0.001			
Propoor								-1.029***		
								(0.249)		
								0.001		
Tenure sec. improve									-3.687***	
									(1.211)	
									0.002	
Combined end use										-19.75
										(38.94)
										0.612
CIDP	-0.097**	0.041	-0.010	-0.061	-0.066	-0.0261	-0.049	-0.056	-0.067	0.079
	(0.042)	(0.0932)	(0.051)	(0.050)	(0.045)	(0.046)	(0.046)	(0.045)	(0.056)	(0.492)
	0.021	0.661	0.841	0.219	0.134	0.566	0.277	0.209	0.229	0.872
Observations	1,225	1,225	1,225	1,225	1,225	1,206	1,206	1,225	1,225	1,225
Countries	19	19	19	19	19	19	19	19	19	19
Other controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CFE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CTE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CTT	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
FS passed	Y	N	Y	Y	Y	Y	Y	Y	P	N

Note: 'Lsrp' stands for years of large-scale rural protests from Albertus (2015). It takes the value of 1 during year/years of protests and 0 otherwise. Only the new control variable, CIDP is reported for expositional purpose. 'CIDP' denotes the active year/years of the clergy initiated democratic participation initiatives including CEBs. 'FS passed' indicate whether the first stage in the IV estimation has been passed or not based on the first stage results reported in Table 3 with Y indicating Yes, N denoting No and P representing Partial, i.e., one F-test is passed. R-squared is not reported in the second stage setup above. Robust standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Table 6. Placebo test with conflict incidence as the outcome variable, first stage results

VARIABLES	(1) LR implemented	(2) Expropriation	(3) Redistributive	(4) Ceiling, expro. & redist.	(5) LR maj-min.	(6) LR national	(7) Propoor	(8) Tenure sec. improve
Elite split	0.136***	0.145***	0.126***	0.135***	0.305***	0.160***	0.138***	0.047***
	(0.023)	(0.028)	(0.023)	(0.023)	(0.052)	(0.034)	(0.023)	(0.011)
	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
CIDP	0.016	0.059**	0.004	0.013	0.188***	0.047	0.032	0.006
	(0.023)	(0.024)	(0.024)	(0.024)	(0.052)	(0.036)	(0.024)	(0.012)
	0.576	0.017	0.867	0.586	0.000	0.190	0.186	0.599
KP rk LM stat	34.133	25.178	28.807	32.444	34.654	23.021	35.056	18.343
[p-val]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
KP Wald rk F stat	34.877	25.187	28.827	32.871	34.284	22.008	36.031	17.706
Cragg-Donald Wald F statistic	54.895	42.523	46.631	52.674	53.534	30.573	57.416	27.210
Observations	956	956	956	956	937	937	956	956
Countries	19	19	19	19	19	19	19	19
Other controls	Y	Y	Y	Y	Y	Y	Y	Y
CFE	Y	Y	Y	Y	Y	Y	Y	Y
CTE	Y	Y	Y	Y	Y	Y	Y	Y
CTT	Y	Y	Y	Y	Y	Y	Y	Y

Note: The dependent variables are various motives of land reforms implemented. We are dropping two land reforms ('Landceiling' and 'Combined end use') from the second stage results reported below as these did not pass the first stage tests. The instrumental variable is the years of elite split from Albertus (2015). Refer to Table 1 for detailed names of the land reform variables. Only the new control variable, CIDP is reported for expositional purpose. 'CIDP' denotes the active year/years of the clergy initiated democratic participation initiatives including CEBs. KP stands for Kleibergen Paap tests. The Stock-Yogo critical value for 10% relative bias is 16.38. Hansen's J-statistic is not reported as the equation is exactly identified with only instrument (elite split). CFE represents country-specific fixed effects, CTE represents time fixed effects, CTT stands for country-specific time trend. Robust standard errors are reported in parentheses; constants from second stage are not reported. *** p<0.01, ** p<0.05, * p<0.

Table 7. Placebo results with conflict incidence as the outcome variable (second stage)

VARIABLES	(1) Conf	(2) Conf	(3) Conf	(4) Conf	(5) Conf	(6) Conf	(7) Conf	(8) Conf	(9) Conf	(10) Conf
LR implemented	0.749***									
	(0.187)									
	0.000									
Landceiling		27.35								
		(80.11)								
		0.733								
Expropriation			0.704***							
			(0.181)							
			0.0001							
Redistributive				0.810***						
				(0.209)						
				0.0001						
Ceiling, expro. & redist.					0.755***					
					(0.192)					
					0.000					
LR maj-min.						0.340***				
						(0.0867)				
						0.000				
LR national							0.647***			
							(0.181)			
							0.0004			
Propoor								0.739***		
								(0.184)		
								0.000		
Tenure sec. improve									2.156***	
									(0.598)	
									0.0003	
Combined end use										5.417
										(4.228)
										0.200
CIDP	0.0409	-0.763	0.0110	0.0496	0.0427	-0.0110	0.0224	0.0294	0.0391	0.00954
	(0.0327)	(2.545)	(0.0355)	(0.0338)	(0.0339)	(0.0397)	(0.0384)	(0.0339)	(0.0385)	(0.0942)
	0.211	0.764	0.757	0.143	0.208	0.782	0.560	0.385	0.310	0.919
Observations	956	956	956	956	956	937	937	956	956	956
Countries	19	19	19	19	19	19	19	19	19	19
Other controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CFE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CTE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CTT	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
FS passed	Y	N	Y	Y	Y	Y	Y	Y	Y	N

Note: The dependent variable (Conf) refers to incidence of conflict which takes the value 1 if there is a conflict threshold achieved in terms of deaths per year in a particular country as a result of ongoing civil strife and 0, otherwise. The variable is in the spirit of Hull and Imai (2013 JDE) and Fearon and Laitin (2003). Please refer to Table 1 for description of all other variable names. 'FS passed' indicate whether the first stage in the IV estimation has been passed or not based on the first stage results reported in Table 10 appendix with Y indicating Yes, N denoting No and P representing Partial, i.e., one F-test is passed. R-squared is not reported in the second stage setup above. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table 8. Mechanism with land redistributed, first stage results

VARIABLES	(1) LR implemented	(2) Expropriation	(3) Redistributive	(4) Ceiling, expro. & redist.	(5) LR maj-min.	(6) LR national	(7) Propoor	(8) Tenure sec. improve
Elite split	0.128***	0.108***	0.094***	0.117***	0.350***	0.181***	0.118***	0.036***
	(0.021)	(0.025)	(0.021)	(0.021)	(0.048)	(0.031)	(0.021)	(0.009)
	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
CIDP	-0.014	0.061**	0.012	0.010	0.191***	0.052	0.027	0.004
	(0.023)	(0.022)	(0.022)	(0.023)	(0.054)	(0.035)	(0.022)	(0.011)
	0.656	0.007	0.578	0.656	0.000	0.139	0.218	0.694
KP rk LM stat	35.256	18.643	20.263	29.963	51.493	34.969	30.979	14.739
[p-val]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
KP Wald rk F stat	34.465	17.894	19.278	29.016	51.947	33.910	30.226	13.902
Cragg-Donald Wald F statistic	53.918	32.075	32.399	48.662	83.493	48.742	50.678	22.212
Observations	1,164	1,164	1,164	1,164	1,163	1,163	1,164	1,164
Countries	19	19	19	19	19	19	19	19
Other controls	Y	Y	Y	Y	Y	Y	Y	Y
CFE	Y	Y	Y	Y	Y	Y	Y	Y
CTE	Y	Y	Y	Y	Y	Y	Y	Y
CTT	Y	Y	Y	Y	Y	Y	Y	Y

Note: The dependent variables are various motives of land reforms implemented. We are dropping two land reforms ('Landceiling' and 'Combined end use') from the baseline Table 3 as these did not pass the first stage tests. Refer to Table 1 for detailed names of the land reform variables. The instrumental variable is the years of elite split from Albertus (2015). Only the new control variable, CIDP is reported for expositional purpose. 'CIDP' denotes the active year/years of the clergy initiated democratic participation initiatives including CEBs. KP stands for Kleibergen Paap tests. The Stock-Yogo critical value for 5% relative bias is 13.91 and 10% relative bias is 9.08. Hansen's J-statistic is not reported as the equation is exactly identified with only instrument (elite split). CFE represents country-specific fixed effects, CTE represents time fixed effects, CTT stands for country-specific time trend. Robust standard errors are reported in parentheses; constants from second stage are not reported. *** p<0.01, ** p<0.05, * p<0.

Table 9. Mechanism with land redistributed, second stage IV results

VARIABLES	(1) Land redist.	(2) Land redist.	(3) Land redist.	(4) Land redist.	(5) Land redist.	(6) Land redist.	(7) Land redist.	(8) Land redist.
LR implemented	2.314***							
	(0.419)							
	0.000							
Expropriation		2.742***						
		(0.586)						
		0.000						
Redistributive			3.143***					
			(0.660)					
			0.000					
Ceiling, expro. & redist.				2.542***				
				(0.466)				
				0.000				
LR maj-min.					0.851***			
					(0.0981)			
					0.000			
LR national						1.647***		
						(0.252)		
						0.000		
Propoor							2.521***	
							(0.457)	
							0.000	
Tenure sec. improve								8.147***
								(2.065)
								0.000
CIDP	0.309***	0.106	0.236***	0.249***	0.112**	0.189***	0.206***	0.238**
	(0.069)	(0.073)	(0.072)	(0.064)	(0.049)	(0.060)	(0.065)	(0.094)
	0.000	0.147	0.001	0.000	0.024	0.001	0.001	0.012
Observations	1,164	1,164	1,164	1,164	1,163	1,163	1,164	1,164
Countries	19	19	19	19	19	19	19	19
Other controls	Y	Y	Y	Y	Y	Y	Y	Y
CFE	Y	Y	Y	Y	Y	Y	Y	Y
CTE	Y	Y	Y	Y	Y	Y	Y	Y
CTT	Y	Y	Y	Y	Y	Y	Y	Y

Note: The dependent variable is the natural logarithm of percentage of land redistributed from Albertus (2015). Refer to Table 1 for detailed names of the land reform variables. Only the new control variable, CIDP is reported for expositional purpose. 'CIDP' denotes the active year/years of the clergy initiated democratic participation initiatives including CEBs. R-squared is not reported for the second stage results shown above. Robust standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Table 10. Mechanism with landgini, first stage results

VARIABLES	(1) LR implemented	(2) Expropriation	(3) Redistributive	(4) Ceiling, expro. & redist.	(5) LR maj-min.	(6) LR national	(7) Propoor	(8) Tenure sec. improve
Elite split	0.137***	0.142***	0.123***	0.132***	0.330***	0.178***	0.141***	0.048***
	(0.024)	(0.029)	(0.024)	(0.025)	(0.053)	(0.035)	(0.024)	(0.011)
	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
CIDP	0.013	0.043*	0.004	0.013	0.180***	0.037	0.027	0.004
	(0.024)	(0.025)	(0.025)	(0.026)	(0.054)	(0.037)	(0.025)	(0.013)
	0.576	0.088	0.854	0.603	0.001	0.317	0.287	0.784
KP rk LM stat	31.494	23.724	25.664	29.376	38.067	27.205	33.162	18.780
[p-val]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
KP Wald rk F stat	31.211	23.075	24.907	28.910	37.619	25.995	33.193	17.973
Cragg-Donald Wald F statistic	50.446	38.089	40.422	46.399	59.916	37.079	54.676	26.602
Observations	923	923	923	923	906	906	923	923
Countries	17	17	17	17	17	17	17	17
Other controls	Y	Y	Y	Y	Y	Y	Y	Y
CFE	Y	Y	Y	Y	Y	Y	Y	Y
CTE	Y	Y	Y	Y	Y	Y	Y	Y
CTT	Y	Y	Y	Y	Y	Y	Y	Y

Note: The dependent variables are various motives of land reforms implemented. We are dropping two land reforms ('Landceiling' and 'Combined end use') from the baseline Table 3 as these did not pass the first stage tests. The instrumental variable is the years of elite split from Albertus (2015). Refer to Table 1 for detailed names of the land reform variables. Only the new control variable, CIDP is reported for expositional purpose. 'CIDP' denotes the active year/years of the clergy initiated democratic participation initiatives including CEBs. KP stands for Kleibergen Paap tests. The Stock-Yogo critical value for 5% relative bias is 13.91 and 10% relative bias is 9.08. Hansen's J-statistic is not reported as the equation is exactly identified with only instrument (elite split). CFE represents country-specific fixed effects, CTE represents time fixed effects, CTT stands for country-specific time trend. Robust standard errors are reported in parentheses; constants from second stage are not reported. *** p<0.01, ** p<0.05, * p<0.

Table 11. Mechanism with landgini, second stage IV results

VARIABLES	(1) Landgini	(2) Landgini	(3) Landgini	(4) Landgini	(5) Landgini	(6) Landgini	(7) Landgini	(8) Landgini
LR implemented	-0.006**							
	(0.002)							
	0.012							
Expropriation		-0.005**						
		(0.003)						
		0.013						
Redistributive			-0.007**					
			(0.003)					
			0.014					
Ceiling, expro. & redist.				-0.006**				
				(0.002)				
				0.013				
LR maj-min.					-0.002**			
					(0.001)			
					0.011			
LR national						-0.004**		
						(0.001)		
						0.013		
Propoor							-0.006**	
							(0.002)	
							0.011	
Tenure sec. improve								-0.017**
								(0.006)
								0.011
CIDP	0.001	0.001	0.000	0.001	0.001	0.001	0.000	0.002
	(0.001)	(0.003)	(0.000)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)
	0.440	0.234	0.541	0.447	0.108	0.273	0.333	0.483
Observations	923	923	923	923	906	906	923	923
Countries	17	17	17	17	17	17	17	17
Other controls	Y	Y	Y	Y	Y	Y	Y	Y
CFE	Y	Y	Y	Y	Y	Y	Y	Y
CTE	Y	Y	Y	Y	Y	Y	Y	Y
CTT	Y	Y	Y	Y	Y	Y	Y	Y

Note: The dependent variables are the landgini values from Frankema (2010) which are interpolated for intermediate years as in Albertus (2013). R-squared is not reported for the second stage results. Refer to Table 1 for detailed names of the land reform variables. Only the new control variable, CIDP is reported for expositional purpose. 'CIDP' denotes the active year/years of the clergy initiated democratic participation initiatives including CEBs. Other controls include one lag of log GDP per capita, log of population, oil dummy as explained in the notes of Table 2, Non-contiguous territory as explained in the notes of Table 2, democracy dummy as explained in the notes of Table 2 and fractionalization measure as explained in the notes of Table 2. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Figures

Figures A. Figures based on event study analysis with large-scale rural protests as the outcome variable

Figure 1. The effect of all land reforms implemented (LR implemented) on large-scale rural protests

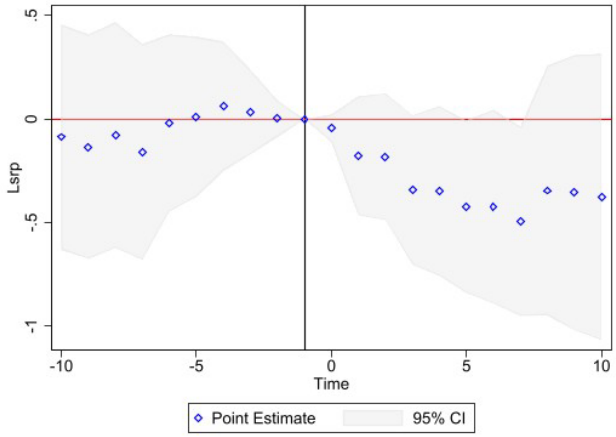


Figure 2. The effect of redistributive land reforms implemented (Redistributive) on large-scale rural protests

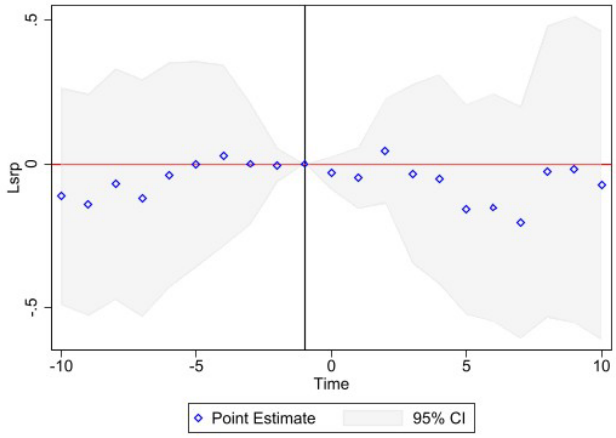


Figure 3. The effect of combined land ceiling, expropriation and redistributive land reforms implemented (Ceiling, expro. & redist.) on large-scale rural protests

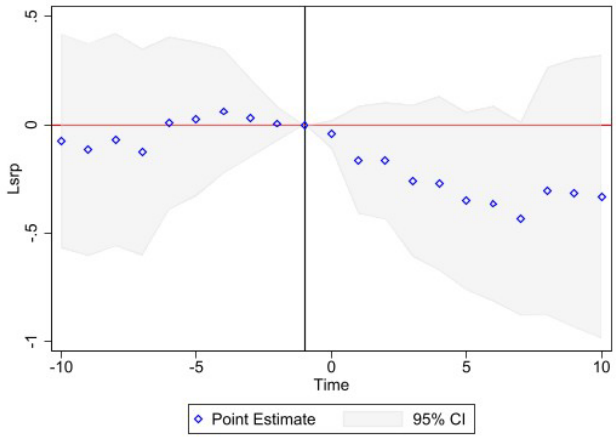


Figure 4. The effect of global major/minor land reform implemented from Albertus (2015) (LR maj-min.) on large-scale rural protests

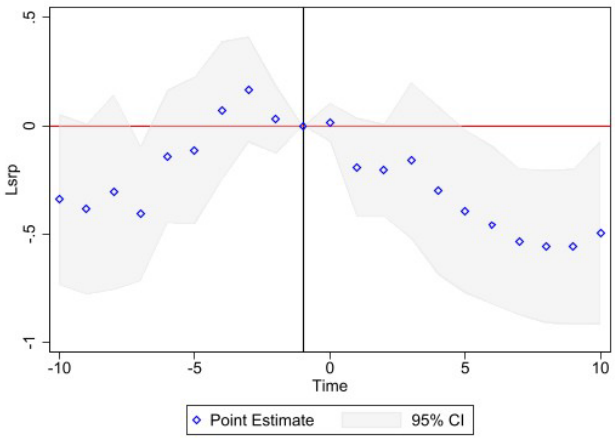


Figure 5. The effect of pro-poor land reform implemented (Propoor) on large-scale rural protests

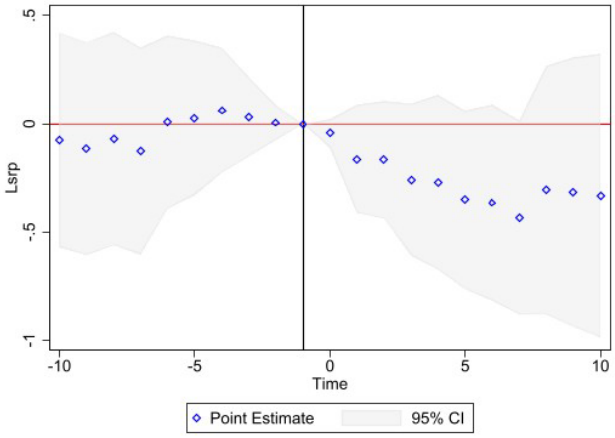
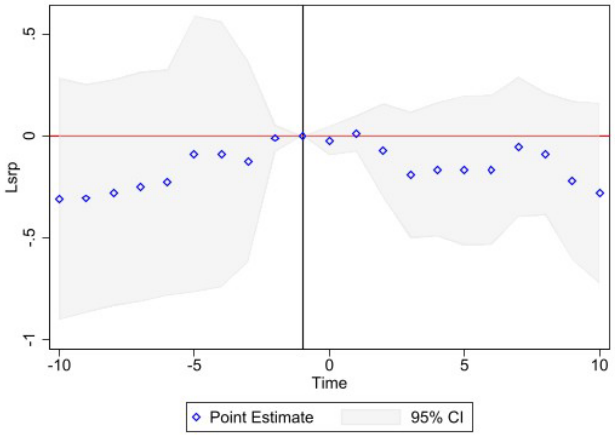


Figure 6. The effect of tenure security improvement land reform implemented (Tenure sec. improve) on large-scale rural protests



Figures B. Figures based on event study analysis with conflict incidence as the outcome variable (placebo check)

Figure 7. The effect of all land reforms implemented (LR implemented) on conflict incidence

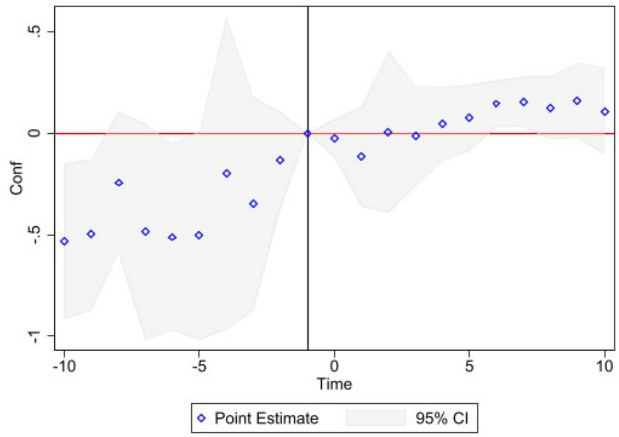


Figure 8. The effect of redistributive land reforms implemented (Redistributive) on conflict incidence

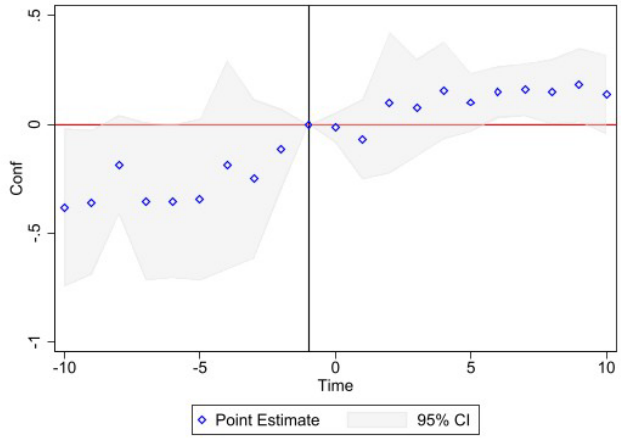


Figure 9. The effect of combined land ceiling, expropriation and redistributive land reforms implemented (Ceiling, expro. & redist.) on conflict incidence

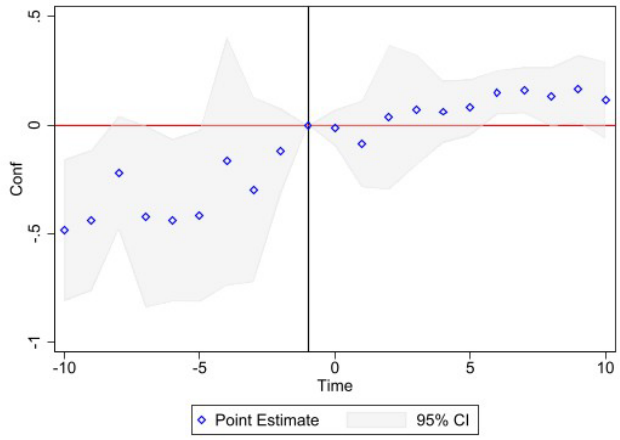


Figure 10. The effect of global major/minor land reform implemented from Albertus (2015) (LR maj-min.) on conflict incidence

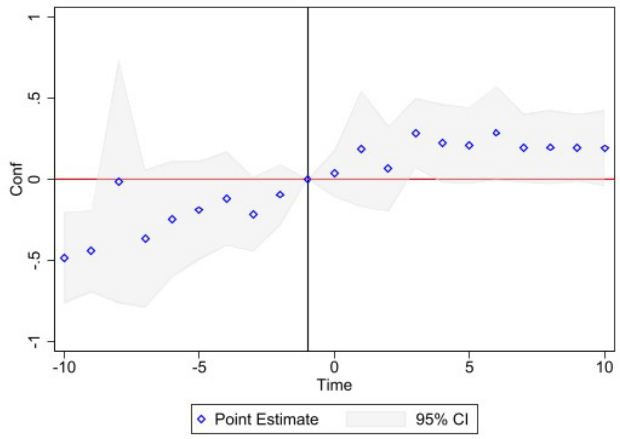


Figure 11. The effect of pro-poor land reform implemented (Propoor) on conflict incidence

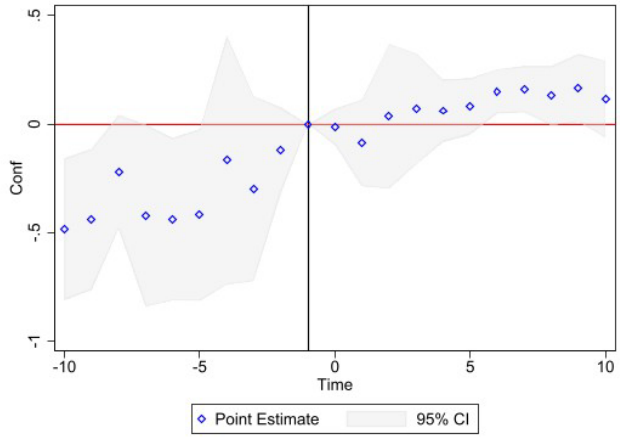
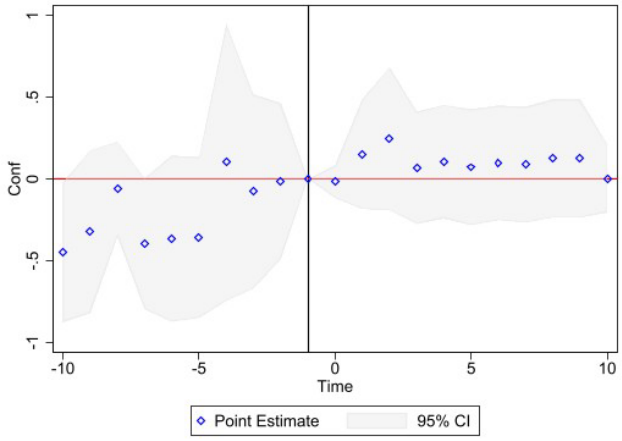


Figure 12. The effect of tenure security improvement land reform implemented (Tenure sec. improve) on conflict incidence



Figures C. Figures based on event study analysis with land redistributed as the outcome variable (first mechanism)

Figure 13. The effect of all land reforms implemented (LR implemented) on percentage of land redistributed

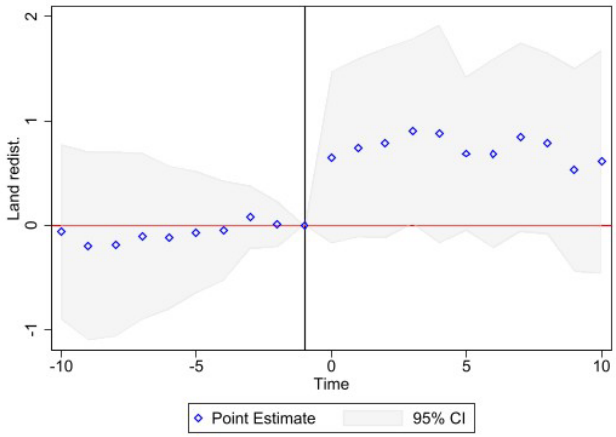


Figure 14. The effect of redistributive land reforms implemented (Redistributive) on percentage of land redistributed

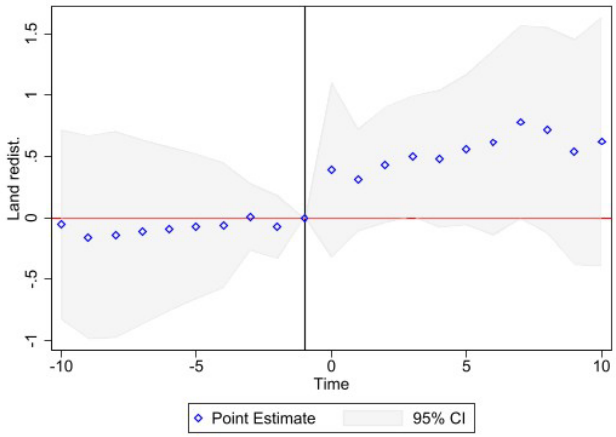


Figure 15. The effect of combined land ceiling, expropriation and redistributive land reforms implemented (Ceiling, expro. & redist.) on percentage of land redistributed

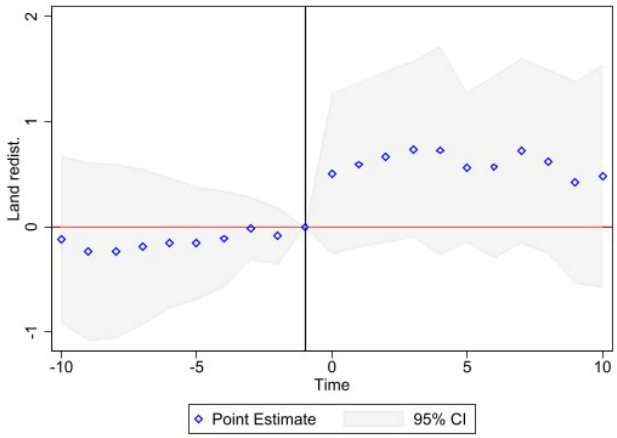


Figure 16. The effect of global major/minor land reform implemented from Albertus (2015) (LR maj-min.) on percentage of land redistributed

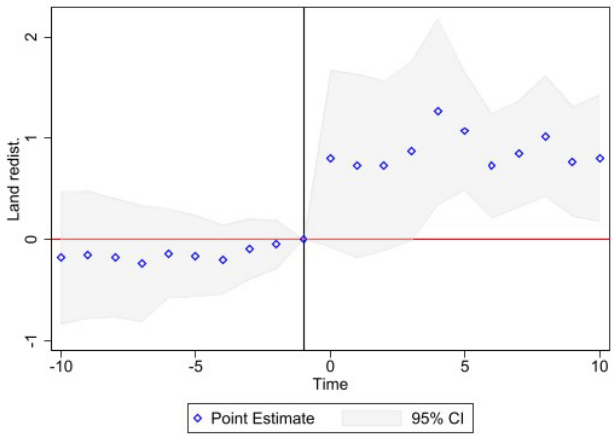


Figure 17. The effect of pro-poor land reform implemented (Propoor) on percentage of land redistributed

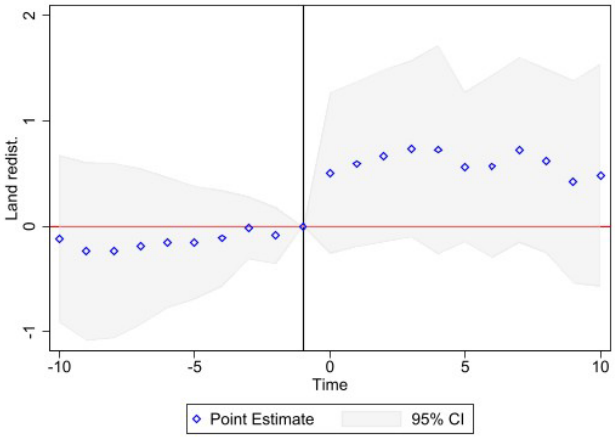
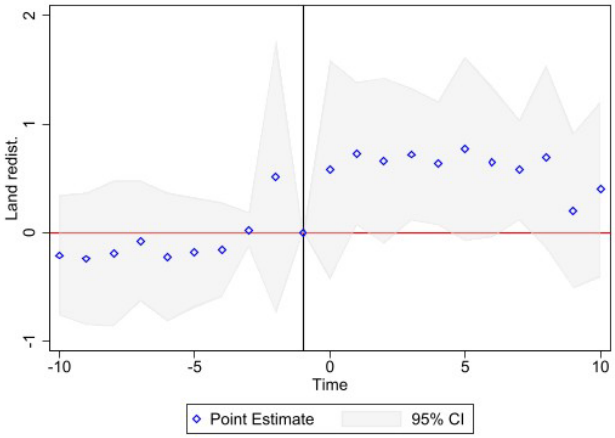


Figure 18. The effect of tenure security improvement land reform implemented (Tenure sec. improve) on percentage of land redistributed



Figures D. Figures based on event study analysis with land gini as the outcome variable (second mechanism)

Figure 19. The effect of all land reforms implemented (LR implemented) on land gini

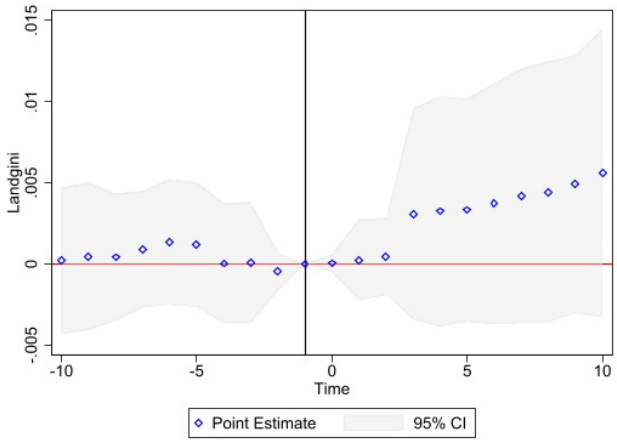


Figure 20. The effect of redistributive land reforms implemented (Redistributive) on land gini

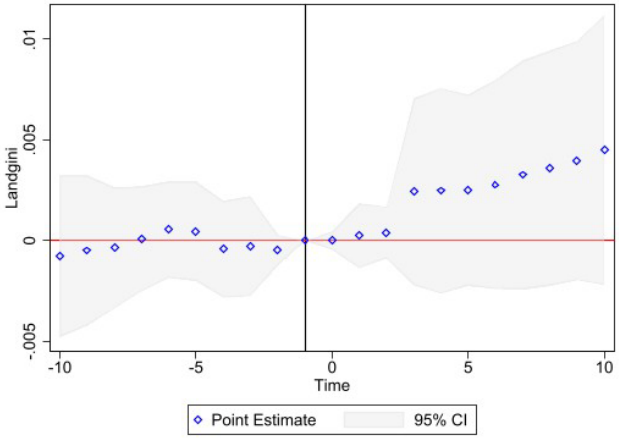


Figure 21. The effect of combined land ceiling, expropriation and redistributive land reforms implemented (Ceiling, expro. & redist.) on land gini

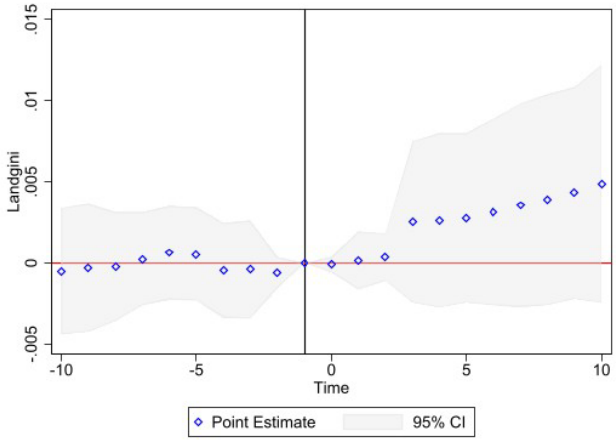


Figure 22. The effect of global major/minor land reform implemented from Albertus (2015) (LR maj-min.) on land gini

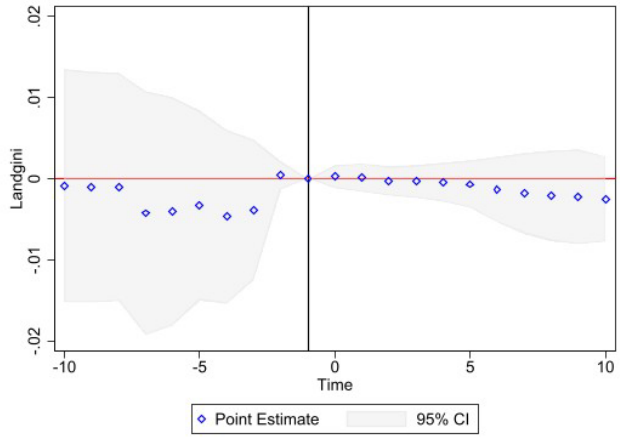


Figure 23. The effect of pro-poor land reform implemented (Propoor) on land gini

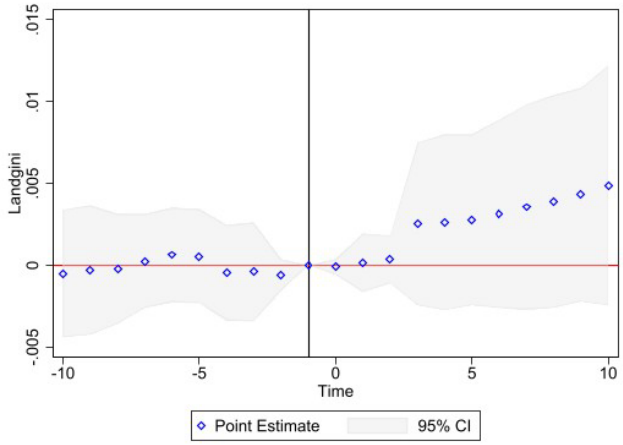
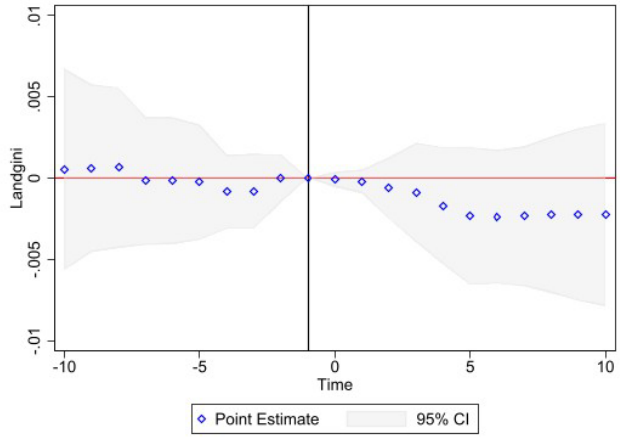


Figure 24. The effect of tenure security improvement land reform implemented (Tenure sec. improve) on land gini



Appendix Tables

Table 2A. Poisson model estimation with fixed effects results, full table with all controls

VARIABLES	(1) Lsrp	(2) Lsrp	(3) Lsrp	(4) Lsrp	(5) Lsrp	(6) Lsrp	(7) Lsrp	(8) Lsrp	(9) Lsrp	(10) Lsrp
LR implemented	-0.0529									
	(0.398)									
	0.894									
Landceiling		0.521								
		(0.746)								
		0.485								
Expropriation			0.177							
			(0.385)							
			0.647							
Redistributive				0.229						
				(0.405)						
				0.572						
Ceiling, expro. & redist.					0.103					
					(0.395)					
					0.793					
LR maj-min.						-0.552***				
						(0.212)				
						0.00919				
LR national							-0.568**			
							(0.285)			
							0.0462			
Propoor								-0.0448		
								(0.385)		
								0.907		
Tenure sec. improve									-0.829**	
									(0.349)	
									0.0175	
Combined end use										0.872**
										(0.435)
										0.0452
CIDP	-0.419*	-0.427*	-0.421*	-0.435*	-0.427*	-0.341	-0.412*	-0.419*	-0.453*	-0.392*
	(0.246)	(0.242)	(0.238)	(0.248)	(0.249)	(0.227)	(0.246)	(0.247)	(0.245)	(0.238)
	0.0887	0.078	0.0775	0.0795	0.0861	0.134	0.0935	0.0899	0.0641	0.0999
GDPl1	0.898	0.982*	1.017	1.054*	0.982	1.112*	1.071*	0.903	0.552	0.802
	(0.643)	(0.592)	(0.643)	(0.626)	(0.630)	(0.584)	(0.594)	(0.635)	(0.590)	(0.532)
	0.162	0.0973	0.114	0.0925	0.119	0.0570	0.0713	0.155	0.349	0.132
Oil	-0.308	-0.305	-0.304	-0.301	-0.302	-0.284	-0.226	-0.306	-0.343	-0.382
	(0.423)	(0.418)	(0.421)	(0.408)	(0.415)	(0.497)	(0.492)	(0.423)	(0.417)	(0.429)
	0.467	0.466	0.470	0.461	0.467	0.568	0.646	0.469	0.411	0.373
Democracy	0.450*	0.429	0.435	0.426	0.435	0.563**	0.583**	0.448*	0.438*	0.417
	(0.271)	(0.282)	(0.277)	(0.279)	(0.276)	(0.280)	(0.283)	(0.272)	(0.266)	(0.285)
	0.0968	0.128	0.116	0.126	0.114	0.0444	0.0394	0.0990	0.0996	0.143
Lpop	-3.198	-2.811	-3.128	-3.177	-3.178	-3.187	-3.356	-3.186	-3.130	-3.111
	(2.191)	(1.919)	(2.087)	(2.111)	(2.142)	(2.039)	(2.100)	(2.192)	(2.195)	(1.940)
	0.144	0.143	0.134	0.132	0.138	0.118	0.110	0.146	0.154	0.109
Observations	1,159	1,159	1,159	1,159	1,159	1,141	1,141	1,159	1,159	1,159
Countries	18	18	18	18	18	18	18	18	18	18
CFE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CTE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CTT	N	N	N	N	N	N	N	N	N	N

Note: 'Lsrp' stands for years of large-scale rural protests from Albertus (2015). It takes the value of 1 during year/years of protests and 0 otherwise. 'LR implemented' denotes all kinds of land reforms implemented as a package; 'LR maj-min.' denotes global major minimum land reform from Albertus (2015). 'LR national' depicts global national land reform from Albertus (2015). Only the new control variable, CIDP is reported for expositional purpose. 'CEB' denotes the active year/years of Base Christian Communities; The unreported controls are, (i) 'GDPl1' is the log per capita GDP; (ii) 'Lpop' is log of population; (iii) 'Oil' is a dummy variable which takes the value 1 if a country earns 1/3 of its exports revenue from oil in that particular year, (iv) 'Democracy' which takes the value 1 if the polity2 score is 5 or above, else 0; (v) 'Nonconti.' which takes the value 1 if the country has non-contiguous territory and (vi) 'Fractionalization' measure from Fearon (2003) paper. We are not reporting estimated coefficients for all other controls and constant in this table (refer to Appendix Table 2ALA for full results). Robust standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1

Table 3A. IV results, second stage, baseline, full result including controls

VARIABLES	(1) Lsrp	(2) Lsrp	(3) Lsrp	(4) Lsrp	(5) Lsrp	(6) Lsrp	(7) Lsrp	(8) Lsrp	(9) Lsrp	(10) Lsrp
LR implemented	-0.911***									
	(0.209)									
	0.000									
Landceiling		-4.390***								
		(1.692)								
		0.009								
Expropriation			-1.200***							
			(0.349)							
			0.001							
Redistributive				-1.309***						
				(0.356)						
				0.002						
Ceiling, expro. & redist					-1.067***					
					(0.262)					
					0.001					
LR maj-min.						-0.317***				
						(0.0838)				
						0.001				
LR national							-0.599***			
							(0.170)			
							0.001			
Propoor								-1.029***		
								(0.249)		
								0.001		
Tenure sec. improve									-3.687***	
									(1.211)	
									0.002	
Combined end use										-19.75
										(38.94)
										0.612
CIDP	-0.0966**	0.0409	-0.0102	-0.0611	-0.0676	-0.0261	-0.0498	-0.0561	-0.0668	0.0794
	(0.0418)	(0.0932)	(0.0507)	(0.0496)	(0.0451)	(0.0455)	(0.0458)	(0.0447)	(0.0555)	(0.492)
	0.0207	0.661	0.841	0.219	0.134	0.566	0.277	0.209	0.229	0.872
GDPl1	-0.412**	-0.900*	-0.727**	-0.589**	-0.452**	0.172*	0.0931	-0.402**	-0.707**	-1.717
	(0.164)	(0.474)	(0.286)	(0.233)	(0.182)	(0.103)	(0.111)	(0.171)	(0.327)	(3.997)
	0.0121	0.0575	0.0109	0.0115	0.0133	0.0951	0.401	0.0185	0.0306	0.667
Oil	-0.237***	-0.0721	-0.140**	-0.227***	-0.232***	-0.125*	-0.0592	-0.227***	-0.277***	0.949
	(0.0663)	(0.0759)	(0.0570)	(0.0776)	(0.0713)	(0.0656)	(0.0757)	(0.0700)	(0.0748)	(2.166)
	0.004	0.342	0.014	0.003	0.002	0.057	0.435	0.001	0.001	0.661
Democracy	0.109***	0.142**	0.0867**	0.0655	0.0877**	0.128***	0.179***	0.0986**	0.0738	-0.290
	(0.0407)	(0.0711)	(0.0430)	(0.0454)	(0.0420)	(0.0400)	(0.0494)	(0.0415)	(0.0516)	(0.817)
	0.007	0.046	0.044	0.149	0.039	0.001	0.000	0.018	0.153	0.722
Lpop	0.616	-1.856*	0.668	0.0814	0.290	-0.00604	-0.0835	0.569	1.169*	8.563
	(0.428)	(1.059)	(0.482)	(0.462)	(0.427)	(0.431)	(0.471)	(0.438)	(0.696)	(15.72)
	0.150	0.079	0.166	0.860	0.498	0.989	0.859	0.194	0.093	0.586
Observations	1,225	1,225	1,225	1,225	1,225	1,206	1,206	1,225	1,225	1,225
Countries	19	19	19	19	19	19	19	19	19	19
CFE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CTE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
CTT	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

Note: R-squared is not reported in the IV setup. Robust standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1. Please see Table 1 notes for description of all variables.

Table 4A. Mechanism with percentage of land distributed, second stage IV results with all controls

VARIABLES	(1) Land redist	(2) Land redist	(3) Land redist	(4) Land redist	(5) Land redist	(6) Land redist	(7) Land redist	(8) Land redist
LR implemented	2.314***							
	(0.419)							
	0.000							
Expropriation		2.742***						
		(0.586)						
		0.000						
Redistributive			3.143***					
			(0.660)					
			0.000					
Ceiling, expro. & redist.				2.542***				
				(0.466)				
				0.000				
LR maj-min.					0.851***			
					(0.0981)			
					0.000			
LR national						1.647***		
						(0.252)		
						0.000		
Propoor							2.521***	
							(0.457)	
							0.000	
Tenure sec. improve								8.147***
								(2.065)
								0.000
CIDP	0.309***	0.106	0.236***	0.249***	0.112**	0.189***	0.206***	0.238**
	(0.069)	(0.073)	(0.072)	(0.064)	(0.049)	(0.060)	(0.065)	(0.094)
	0.000	0.147	0.001	0.000	0.024	0.001	0.001	0.012
GDPl1	1.878***	2.200***	2.204***	1.874***	0.469***	0.674***	1.807***	2.124***
	(0.348)	(0.501)	(0.446)	(0.349)	(0.112)	(0.155)	(0.334)	(0.543)
	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Oil	0.201**	-0.0327	0.171	0.181**	-0.0701	-0.249**	0.174*	0.282**
	(0.083)	(0.074)	(0.111)	(0.092)	(0.069)	(0.103)	(0.091)	(0.120)
	0.016	0.660	0.123	0.050	0.309	0.016	0.056	0.018
Democracy	-0.118**	-0.043	-0.006	-0.062	-0.148***	-0.281***	-0.085	0.003
	(0.057)	(0.060)	(0.063)	(0.057)	(0.042)	(0.064)	(0.058)	(0.083)
	0.040	0.477	0.924	0.275	0.000	0.000	0.139	0.964
Lpop	-0.177	0.257	1.087	0.620	1.796***	2.030***	0.001	-2.107*
	(0.771)	(1.000)	(0.933)	(0.751)	(0.464)	(0.635)	(0.778)	(1.184)
	0.818	0.797	0.244	0.409	0.000	0.001	0.999	0.075
Observations	1,164	1,164	1,164	1,164	1,163	1,163	1,164	1,164
Countries	19	19	19	19	19	19	19	19
CFE	Y	Y	Y	Y	Y	Y	Y	Y
CTE	Y	Y	Y	Y	Y	Y	Y	Y
CTT	Y	Y	Y	Y	Y	Y	Y	Y

Note: R-squared is not reported for the second stage results shown above. Robust standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1. Please refer to Table 1 for description of variable names in the table above.

Data Appendix

This data appendix provides background information and coding the timeline of the new control variable, Church or clergy induced democratic participation including establishment of Christian Base Communities (CEBs).

Background information on Base Christian Communities

The Encyclopaedia mentions that 'Christian Based Communities', one of the ground forces based on the liberal theology showed some heterogeneities across different dioceses in LA. EN reports the following "*The growth of CEBs has been explosive in some countries and virtually nonexistent in others. In still other countries there are great variations among dioceses. The data suggest two critical variables. First, there appears to be a correlation between the extent of poverty in a country and the number of CEBs. CEBs are few and far between in Argentina, Uruguay, Chile, and Costa Rica, countries that have enjoyed a relatively higher standard of living than the rest of Latin America. Second, there is a clear and even stronger correlation between the support a bishop gives the development of CEBs in his diocese and the number of CEBs that are organized. Brazil, with well over a hundred thousand CEBs, best illustrates both variables: widespread poverty and the most progressive bishops of any national church in the world. At the other end of the theological spectrum, Colombia, with equally widespread poverty and one of the most conservative hierarchies in Latin America, has relatively few CEBs. El Salvador, Guatemala, and Mexico also qualify on both points, but the development of CEBs in each of these countries has been highly erratic: in dioceses with progressive bishops large numbers of CEBs flourish, while dioceses headed by conservatives have few.*

It should not be surprising that conservative episcopal reaction echoes in the larger society. CEBs have come to be regarded as subversive by elites bent on maintaining the economic and political status quo. Dozens of priests and nuns involved with CEBs have been harrassed, exiled, tortured, or murdered. Hundreds of catechists and delegates and even some bishops have met the same fate.

Scholars have begun to assess the impact of over two decades of CEB activity, amid the explosion of evangelical Protestantism and Catholic Charismatic Renewal (CCR), whose practitioners were initially known as Pentecostal Catholics. Evidence shows a decrease in CEBs, with the poor in Latin America offering only limited support for liberation theology. Nevertheless, the Conference of Latin American Bishops (CELAM) has called for greater CEB growth, and in some regions, such as the Amazon, CEBs remain vibrant and significant. Representing a significant development in the history of the Catholic Church, CEBs endure in the twenty-first century's competitive religious marketplace."

It appears that the CEBs also capture the difference between Churches allegiances to elite' vs masses (Levine, 1979). In the beginning of the 20th century, i.e., early 1900s, the Church in Latin America in general were engaged with elites and highly influential people in the society (as mentioned by the vast religion and politics papers in the extant literature). The Church was given ample land by the elites and the relationship between clergy and rich people was quite cosy (Harrington, 1977). However, in between 1962 – 1965, based on the Second Vatican Council declaration, the Catholic Church shifted focus on the plight of the poor especially in Latin America and espoused a 'middle way', denouncing both capitalism and socialism (CIA document 1986, Jordan 2022). This created an important shift in terms of

Church focus from the rich to poor. The CEBs served similar to local participatory councils where poor individuals and masses would come together at least a week for '*prayers, discussions and action*' (CIA 1986 document, Jordan 2022, Bruneau, 1980).

The CEBs undertook a number of different political activities apart from offering prayers such as 'human rights monitoring, organizing labor protests, establishing cooperatives and credit unions, providing health care and education, fundraising, supporting solidarity campaigns, publishing and other reform activities' (CIA document 1986). Thus, establishment of CEBs could be seen as an impetus to big ticket reforms like land reforms which are offered as packages with multiple motives associated with it.

However, there are instances where establishment of CEBs do not transfer to more participatory actions in the ground. As the EB document pointed out earlier, the effectiveness of CEBs depend a lot on the local bishops in the dioceses in which they were operating. If the bishops were more conservatives, then CEBs were not that effective to usher in changes in the political dimension, such as in Colombia. In fact, Church in Colombia was state supported and remained quite conservative throughout its' history (Harrington, 1977). Crahan (1979) also mentions the same for Cuban Catholic churches before the revolution where the church emphasis was to maintain good relationship with the elite (see also Levine 1973). On the other hand, bishops in Brazil were more progressive, and the CEBs in Brazil were quite active as a consequence. Similarly, bishops in Chile were considered one of the most progressive in Latin America (Harrington, 1977)

The CEBs also differ in terms of their timeline of effectiveness as the CIA document pointed out that the Catholic Church in April 1986 in its 'Instruction on Christian Freedom and Liberation' mentioned that Church should focus on the plight of the poor, but the active political push by the clergy were not encouraged. Thus, it could be assumed that the explicit focus of the CEBs could have been diverted afterwards in the 1990s. The EB also mentioned that the influence of CEBs were erratic over time, which means it could be very active over some time period, and then there may be a lull in these activities before being rejuvenated at a later point in time. This is another dimension of heterogeneity in the activities of EB which will be addressed in the data coding. This means, some years will have the values 1, then 0, and then again 1.

Coding decision and country specific information

Coding decision

We code the church induced democratic participation based on the following rules. First, the clergy-initiated participation should be documented in at least two academic, peer reviewed journal publications or books. Second, this process should be corroborated by other publications like government reports, respected think tank reports, PhD or Masters' thesis from respected universities etc. Third, the encouragement of the clergy like using radio talks or writing in popular press or the nature or attributes of church participation should be clearly documented in the source materials.

Country specific information

Argentina

The CIDP start and end years are between 1975 to 1977 based on Smith (1979) and Dodson (1979). Smith (1979) mentions that in Argentina, the effective CEBs were not established by the clergy for a longer time period possibly due to the violent repercussions of the dictatorial regime operating at that time. Dodson (1979) also highlights the liberation theology and its ramifications in Argentina in particular and the Latin American region in general. Dodson (1979) writes, "*Today, Latin American bishops and Christian base communities throughout the continent are deeply involved in the struggle to preserve human rights against the encroachments of authoritarian regimes. One of the most controversial aspects of changing Latin American church has been the emergence of organized movements of Christian radicals who sought to use religion as a base from which to transform society through political action. Sizeable priest movements of the left appeared in countries such as Argentina, Chile, Colombia and Peru, where they had a notable impact on social politics.*"

References

Dodson, Michael. 1979. Liberation Theology and Christian Radicalism in Contemporary Latin America. *Journal of Latin American Studies*, 11(1), 203-222.

Smith, Brian H. 1979. Churches and Human Rights in Latin America Recent Trends in the Subcontinent. *Journal of Interamerican Studies and World Affairs*, 21(1), 89-128.

Belize

We do not find any evidence of CIDP in Belize. In particular the following paragraph from US Country Studies on Belize reports the following:

Liberal political movements, such as liberation theology, had not taken root in Belize, and the Roman Catholic Church avoided the split between the so-called "traditional" and "popular" churches that divided Roman Catholics in other Central American countries. Moreover, politicians probably overestimated the ability of the Roman Catholic Church to respond as a monolithic institution, and their perception of so-called "Roman Catholic" positions often lacked an awareness of current Roman Catholic thought and practice. The generally conservative outlook of the Belizean Protestant churches, which shared the traditional Roman Catholic position on many moral and social issues, perhaps reinforced politicians' consciousness of religious interests.

Hence all years for CIDP are coded as 0.

References

US Country Studies for Belize, Information on "Churches and Religious Institutions"

http://countrystudies.us/belize/81.htm

Brazil

The start time and end time for CIDP for Brazil is between 1964 and 1986 based on the CIA documentation (1986) narrative supplemented by information from *Encyclopedia Britannica*.

Bruneau (1980) also provided extensive documentation about the CIDP activities in Brazil. In his paper's abstract, Bruneau (1980) writes:

"The socio-political orientation of the Catholic Church varies and currently in Latin America it has formally assumed a position supporting change and favouring the poor. For historical and structural reasons, the national churches lack influence and are poorly equipped to relate to the poor. Due to the perceived need for new structures and relying on the legitimation of the Church as a transnational actor, the Basic Christian Community (CEB) is promoted as the most appropriate entity for relating to the lower classes. Where the CEBs assume their own dynamic, as in Brazil, they may well invert the structure and orientation of the Church. Through field study and documentation, it is concluded that CEBs hold serious implications for development as the lower classes are participating and demanding changes in the established order."

References

Bruneau, Thomas C. 1980. The Catholic Church and Development in Latin America: The Role of the Basic Christian Communities. *World Development*, 8, 535-544.

CIA Document. 1986. Liberation Theology: Religion, Reform and Revolution A Research Paper, GI 86-10028, Directorate of Intelligence, Central Intelligence Agency, USA.

Encyclopedia of Latin American History and Culture, Encyclopedia.com; <https://www.encyclopedia.com/humanities/encyclopedias-almanacs-transcripts-and-maps/christian-base-communities>

Bolivia

The start and end years of CIDP (1973 – 1977) are based on Smith (1979) article about Bolivia. Note that Smith (1980) mentions that bishops and clergies were not very active in setting up CEBs. However, there are some evidences of helping the masses by encouraging to participate in active land tilling activities as documented in Burke (1970).

References

Burke, M. 1970. Land Reform and its Effect upon Production and Productivity in the Lake Titicaca Region. *Economic Development and Cultural Change*, 18(3), pp. 410-450.

Smith, Brian H. 1979. Churches and Human Rights in Latin America Recent Trends in the Subcontinent. *Journal of Interamerican Studies and World Affairs*, 21(1), 89-128.

Chile

The start and end time of CIDP (1970 to 1985) is based on the narrative from the CIA document (1986). Levine (1979) provides strong support to the narrative and discusses in detail the evolution of CIDP driven activities in Chile.

References

CIA Document. 1986. Liberation Theology: Religion, Reform and Revolution A Research Paper, GI 86-10028, Directorate of Intelligence, Central Intelligence Agency, USA.

Levine, Daniel H. 1979. Religion and Politics, Politics and Religion: An Introduction, *Journal of Interamerican Studies and World Affairs*, 21(1), 5-29.

Colombia

Harrington (1977) presents an illustrious narrative of Catholic Church in Colombia and showed how it was supporting the state or the federal government for a long period of time. This created some strife within the clergy who vouched for conservatism versus libertarians, such as Camilio Torres, who ultimately left the Church and joined the guerrilla group. There are individual instances of clergies trying to help poor people and the marginalized like setting up the 'Minuto de Dios', a model for urban community living in Bogota in 1950s (Harrington, 1977); however, there are dearth of active CEBs unlike in other countries. We code the start year in 1961 and end year in 1969 based on the narrative in Harrington (1977) and Sanders (1982).

References

Harrington, Ronnie Glatz. 1977. The political posture of the Roman Catholic Church in Latin America with specific reference to Colombia and Chile. Masters' Thesis, University of Massachusetts Amherst, USA.

Sanders, Thomas G. (1982). The Politics of Catholicism in Latin America. *Journal of Interamerican Studies and World Affairs*, 24(2), pp. 241-258.

Costa Rica

Costa Rica is a unique case in the sense that there were not very active CIDPSs, which could be attributed to relative political regime stability and prosperity in Costa Rica (*Encyclopedia* document), but importantly as Brett (1993) pointed out that during the tenure of Archbishop Sanabria (1940-1952), there was close coordination between the government, church and the Communist Party, which created a novel programme of social legislation. Sanabria (1940 – 1952) also supported labor union activities, based on this information we are coding the CIDP years as between 1940 to 1952.

References

Brett, Edward T. 1993. The Impact of Religion in Central America: A Bibliographical Essay. *The Americas*, 49(3), pp. 297-341.

Encyclopedia of Latin American History and Culture, Encyclopedia.com; <https://www.encyclopedia.com/humanities/encyclopedias-almanacs-transcripts-and-maps/christian-base-communities>

Cuba

There is not much evidence of CIDP in Cuba. We code the active years as in 1962 and again between 1970 to 1972 based on Crahan (1979). Crahan (1979) points that for Cuban Catholic churches before the revolution were maintaining good relationship with the elite (Levine 1979) and hence, there are not much emphasis on CIDP in Cuba. Jordan (2022) also mentions that the CIPDs are prominent in other countries of the Latin America and not Cuba.

References

Crahan, Margaret E. 1979. Salvation Through Christ or Marx: Religion in Revolutionary Cuba. *Journal of Interamerican Studies and World Affairs*, 21(1), 156-184.

Levine, Daniel H. 1979. Religion and Politics, Politics and Religion: An Introduction, *Journal of Interamerican Studies and World Affairs*, 21(1), 5-29.

Jordan, Brendan. 2022. Christian Base Community movement in the *Database of Religious History*, University of British Columbia, Canada. https://religiondatabase.org/browse/910/#/

Dominican Republic

The source report by Davidson (2018) did not mention anything about CIDPs in Dominican Republic. There are some political partitions in the 1990s, but the issues were not related to CEBs. Thus, it seems activity years would be 0. However, note that the article by Krämer and Vellguth (2013) mention that first steps towards establishing CEBs were taken just after 1950s in Santiago de los Caballeros, the second largest city in Dominican Republic. Accordingly, we code the initial start and end years as 1960 and 1961, and again in 1965.

We also coded the years between 1970 and 1980 as another set of CIDP start and end years as the US Country Studies Religion narrative mentions that Bishops issued various statements supporting poor people and masses during this time interval.

References

Davidson, Christina C. 2018. Analysis of Evangelical Christianity in the Dominican Republic. Research and Ideas Series, History. Global Foundation for Democracy and Development, New York, USA.

Krämer, Klaus and Vellguth, Klaus (eds.). 2013. Small Christian Communities: Fresh Stimulus for a Forward-looking Church. One World Theology series, vol. 2, Claretian Communications Foundation, Inc., Quezon City, Philippines.

US Country Studies, Dominican Republic, narrative on "Religion"; http://countrystudies.us/dominican-republic/33.htm

Ecuador

The CIDP start and end years are between 1962 to 1976. The timeline entries are based on Bishop Proano's activities in setting up CEBs in Ecuador as mentioned in Proctor (2012)'s Master of Philosophy thesis as significant projects which subsequently helped the masses in Ecuador. It is also supported by the narrative from the *Encyclopedia of Latin American History and Culture*. Krämer and Vellguth (2013) points that (Footnote 434) steps towards setting up CEBs as part of CIDPs have been undertaken during this time period.

References

Krämer, Klaus and Vellguth, Klaus (eds.). 2013. Small Christian Communities: Fresh Stimulus for a Forward-looking Church. One World Theology series, vol. 2, Claretian Communications Foundation, Inc., Quezon City, Philippines.

Proctor, Gerald. 2012. A Commitment to Neighbourhood Base Ecclesial Communities in Global Perspective. *Master of Philosophy Thesis*, Liverpool Hope University, UK.

Encyclopedia of Latin American History and Culture, Encyclopedia.com; <https://www.encyclopedia.com/humanities/encyclopedias-almanacs-transcripts-and-maps/christian-base-communities>

El Salvador

The starting year of CIDP is 1974 and the end point is 1985 based on the CIA document (1986). The end time is 1985 as the Archbishop in 1980 started questioning the participation of Church clergy in insurgent activities. In 1985, the Church came out in support of the Government in August 1985 as per CIA document (1986) and we treat this as the end point of CIDP. Brett (1993) corroborates the above finding with extensive description of church induced democratic participation activities in El Salvador.

References

Brett, Edward T. 1993. The Impact of Religion in Central America: A Bibliographical Essay. *The Americas*, 49(3), pp. 297-341.

CIA Document. 1986. Liberation Theology: Religion, Reform and Revolution A Research Paper, GI 86-10028, Directorate of Intelligence, Central Intelligence Agency, USA.

Guatemala

The start and end point of CIDP is between 1970 till 1986 based on the CIA document (1986). This is supplemented by information from the *Encyclopedia* about the military regimes and their interaction with the clergy. Burnett (1989) provides supporting documentation for CIDP activities in Guatemala.

References

Burnett, Garrard. 1989. Protestantism in Rural Guatemala, 1872-1954. *Latin American Research Review*, 127-142.

CIA Document. 1986. Liberation Theology: Religion, Reform and Revolution A Research Paper, GI 86-10028, Directorate of Intelligence, Central Intelligence Agency, USA.

Encyclopedia of Latin American History and Culture, Encyclopedia.com; <https://www.encyclopedia.com/humanities/encyclopedias-almanacs-transcripts-and-maps/christian-base-communities>

Honduras

We have coded Honduras entries for CIDPs from Shepherd (1993). Honduras experience was quite unique as one of the progressive bishops, Mr. Evelio Dominguez started active rural participation via 'radio talks' from 1961 onwards. Hence, the CIDP variables takes the value 1 from 1960 onward till 1975. Edward and Brett (1988) document how the church devised a model social action programme in 1965 and continued until the Horcone massacre in 1975. In the same vein, White (1977) sheds lights on the church initiated social development programmes and their effects during this time period.

References

Edward, T and Brett, Donna W. 1988. Facing the Challenge: The Catholic Church and the Social Change in Honduras. in *Central America: Historical Perspectives on the Contemporary Crises (Contributions to the Study of World History)*, edited by Ralph Lee Woodward Jr., Greenwood Press, Westport, CT, USA.

Shepherd, Frederic M. 1993. Church and State in Honduras and Nicaragua Prior to 1979. *Sociology of Religion*, 54(3), pp. 277-293.

White, Robert A. 1977. Structural Factors in Rural Development: The Church and the Peasant in Honduras. PhD Dissertation, Cornell University, Ithaca, USA.

Nicaragua

The starting point of CIDP is 1965 based on the information and narrative provided in the CIA document (1986). The timeline coincides with the Liberal Theology movement experienced in Nicaragua between 1965 and 1986. The end point of CIDP is 1990 with the last four entries based on the CIDP narrative from Harrington (1977). Gunderson (2018) provides fresh perspective of CIDP in Nicaragua focusing on the old CEBs and new CEBs.

References

CIA Document. 1986. Liberation Theology: Religion, Reform and Revolution A Research Paper, GI 86-10028, Directorate of Intelligence, Central Intelligence Agency, USA.

Gunderson, Lara M. 2018. Relanzamiento of Nicaragua's Christian Base Communities: Forging New Models of Church and Society for the Twenty-First Century. *Doctoral Thesis, Anthropology*, The University of New Mexico, Albuquerque, New Mexico, USA.

Harrington, Ronnie Glatz. 1977. The political posture of the Roman Catholic Church in Latin America with specific reference to Colombia and Chile. *Masters' Thesis*, University of Massachusetts Amherst, USA.

Mexico

The CIDP start and end years are between 1980 and 1995. The timeline is based on Anderson and Colombo (1988) 's narrative, US Country studies Mexico region "Religious" section's documentation and information provided in the MacNabb and Rees (1993) paper.

References

Anderson, Joan B. and Colombo, J. A. 1988. Christian Base Communities and Grass-roots Development. *The Journal of Behavioral Economics*, 17(2), 97-112.

MacNabb, Valerie A. and Rees, Martha W. 1993. Liberation or Theology? Ecclesial Base Communities in Oaxaca, Mexico. *Journal of Church and State*, 35(4), 723-749.

US Country Studies, 1996. Mexico, section on 'Religion'. http://countrystudies.us/mexico/61.htm

Panama

We code the years of CIDP between 1965 and 1980 based on the Anderson and Colombo (1988) article's narrative about the CIDPs in Panama. The timeline is also supported by Krämer and Vellguth (2013).

References

Anderson, Joan B. and Colombo, J. A. 1988. Christian Base Communities and Grass-roots Development. *The Journal of Behavioral Economics*, 17(2), 97-112.

Krämer, Klaus and Vellguth, Klaus (eds.). 2013. Small Christian Communities: Fresh Stimulus for a Forward-looking Church. One World Theology series, vol. 2, Claretian Communications Foundation, Inc., Quezon City, Philippines.

Paraguay

The CIDP start and end years (1965 – 1980) are based on the narrative from Smith (1979). The entries are based on Smith (1979) article where it says that the Church started helping peasants and poor by setting up credit unions etc in mid-1960s. The CEBs were developed after 1976 crackdown by the government forces. Bruneau (1980) also mentions that there are "several hundred" of CEBs in Paraguay.

References

Bruneau, Thomas C. 1980. The Catholic Church and Development in Latin America: The Role of the Basic Christian Communities. *World Development*, 8, 535-544.

Smith, Brian H. 1979. Churches and Human Rights in Latin America Recent Trends in the Subcontinent. *Journal of Interamerican Studies and World Affairs*, 21(1), 89-128.

Peru

The CIDP start and end years are between 1965 till 1980. These timeline entries are based on Peña (1994) article which mentions in detail the establishments of CEBs in Peru. The CIA document (1986) also refers to active church-initiated protests in Peru in relation to government restrictions on political freedoms. In addition, the CIA document (1986) explicitly states that in Peru, the effective Christian base community networks are being developed around the timeline mentioned above.

References

CIA Document. 1986. Liberation Theology: Religion, Reform and Revolution A Research Paper, GI 86-10028, Directorate of Intelligence, Central Intelligence Agency, USA.

Peña, Milagros. 1994. Liberation Theology in Peru: An Analysis of the Role of Intellectuals in Social Movements. *Journal for the Scientific Study of Religion*, 33(1), 34-45.

Uruguay

The CIDP start and end years are between 1973 to 1975. These entries are based on Edmonds (2014) paper. She mentions ineffectiveness of Catholic Church in criticizing the military rulers and developing CEBs based on a number of reasons. Bruneau (1980) echoes the same sentiment by highlighting that Catholic Churches in Uruguay did not speak against the military regimes.

References

Bruneau, Thomas C. 1980. The Catholic Church and Development in Latin America: The Role of the Basic Christian Communities. *World Development*, 8, 535-544.

Edmonds, Amy. 2014. Moral Authority and Authoritarianism: The Catholic Church and the Military Regime in Uruguay. *Journal of Church and State*, 56(4), 644-669.

Venezuela

Levine (1976) illustrates how Catholic Church in Venezuela was influenced by the elites and was in line with the political regimes all the time. In that sense, the activism associated with setting up of CEBs and maintain such organizations were virtually non-existent in Venezuela. Thus, we code only two years of potential CIDP activities in Venezuela, 1970 and 1971 based on Levine (1976). Levine (1979) reaffirms the above finding in a later study.

References

Levine, Daniel H. 1976. Democracy and the Church in Venezuela, *Journal of Interamerican Studies and World Affairs*, 18(1), 3-23.

Levine, Daniel H. 1979. Church Elites in Venezuela and Colombia: Context, Background and Belief's, Latin American Research Review, 14(1), 51-79.